Control Systems

Previous Year Questions • 1992–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Control Systems (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry

> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

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UNIT

I

Basics of Control Systems

Topics

1. Introduction to Control Systems
2. Laplace Transform and Concept of a System
3. Transfer Function
4. Concept of Feedback, OLTF and Sensitivity
5. Poles, Zeros and Final Value Theorem
6. Dominant Pole, Impulse and Step Response

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Unit Snapshot*

Stream: IN	**Questions:** 23	**MCQ:** 21	**MSQ:** 0	**NAT:** 2
1M: 14	**2M:** 7	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2024)–2M(2017)

**See preface for how Avg Weightage and Range are calculated.*

Q1.24.1 MCQ 2024 1M

The pressure drop across a control valve is constant. The control valve with inherent characteristic has decreasing sensitivity. If x represents the fraction of maximum stem position of the control valve, then the function $f(x)$ representing the fraction of maximum flow is

A. α^{x-1}, where α is constant **B.** $\sqrt{x}$
C. x **D.** x^2

Q1.22.1 MCQ 2022 1M

A unity-gain negative-feedback control system has a loop-gain $L(s)$ given by : $L(s) = \dfrac{6}{s(s-5)}$. The closed-loop system is

A. Causal and stable **B.** Causal and unstable
C. Non-causal and stable **D.** Non-causal and unstable

Q1.19.1 MCQ 2019 1M

In a cascade control system, the closed loop transfer function of the inner loop may be assumed to have a single time-constant τ_1. Similarly, the closed loop transfer function of the outer loop may be assumed to have a single time-constant τ_2. The desired relationship between τ_1 and τ_2 in a well-designed control system is

A. τ_1 is much less than τ_2

B. τ_1 is equal to τ_2

C. τ_1 is much greater than τ_2

D. τ_1 is independent of τ_2

Q1.17.1 MCQ 2017 1M

The term hysteresis is associated with

A. ON-OFF control **B.** P-I control
C. Feed-forward control **D.** Ratio control

Q1.17.2 MCQ 2017 1M

The standard for long distance analog signal transmission in process control industry is

A. 4 – 20 mV **B.** 0 – 20 mA
C. 4 – 20 mA **D.** 0 – 5 V

Q1.16.1 MCQ 2016 2M

The relationship between the force $f(t)$ and the displacement $x(t)$ of a spring-mass system (with a mass M, viscous damping D and spring constant K) is

$$M\frac{d^2x(t)}{dt^2} + D\frac{dx(t)}{dt} + Kx(t) = f(t).$$

$X(s)$ and $F(s)$ are the Laplace transforms of $x(t)$ and $f(t)$ respectively. With $M = 0.1$, $D = 2$, $K = 10$ in appropriate units, the transfer function $G(s) = \dfrac{X(s)}{F(s)}$ is

A. $\dfrac{10}{s^2 + 20s + 100}$ **B.** $s^2 + 20s + 100$

C. $\dfrac{10s^2}{s^2 + 20s + 100}$ **D.** $\dfrac{s}{s^2 + 20s + 100}$

Q1.10.1 MCQ 2010 2M

The rotor of the control transformer of a synchro pair gives a maximum voltage of 1.0 V at a particular position of the rotor of the control transmitter. The transmitter rotor is now rotated by 30° anticlockwise keeping the transformer rotor stationary. The transformer rotor voltage for this position is

A. 1.0 V **B.** 0.866 V
C. 0.5 V **D.** 0 V

Q1.07.1 MCQ 2007 1M

A feedback control system with high gain K, is shown in the figure below:

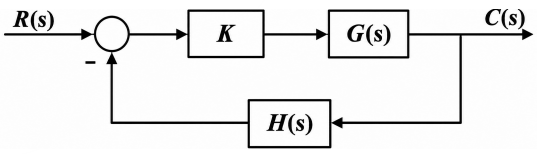

Fig. Q1.07.1

Then the closed loop transfer function is

A. sensitive to perturbations in $G(s)$ and $H(s)$

B. sensitive to perturbations in $G(s)$ but not to perturbations in $H(s)$

C. sensitive to perturbations in $H(s)$ but not to perturbations in $G(s)$

D. insensitive to perturbations in $G(s)$ and $H(s)$

Q1.06.1 MCQ 2006 1M

Compared to a conventional two-phase induction motor, a two-phase servomotor has

A. a smaller R/X ratio and a lighter rotor

B. the same R/X ratio and a heavier rotor

C. a larger R/X ratio and a heavier rotor

D. a larger R/X ratio and a lighter rotor

Q1.05.1 MCQ 2005 1M

A control loop structure is shown in the figure with FC, FT, TC and TT denoting Flow Controller, Flow Transmitter, Temperature Controller and Temperature Transmitter respectively. This loop is an example of

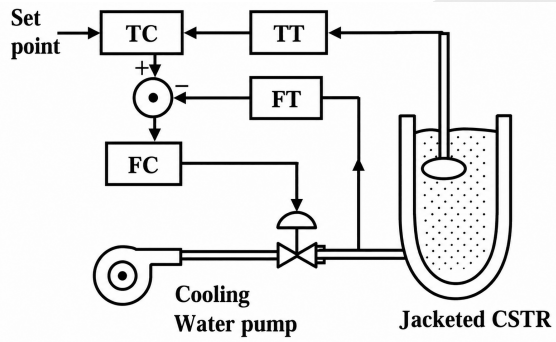

Fig. Q1.05.1

A. Ratio Control **B.** Split Range Control
C. Feed forward Control **D.** Cascade Control

Q1.05.2 MCQ 2005 2M

A single stack stepper motor has 4-stator slots and 18 permanent magnet type rotor pole faces. The step size under half stepping scheme is

A. 5° **B.** 15° **C.** 10° **D.** 20°

Q1.04.1 MCQ 2004 1M

The dominant poles of a servosystem are located at $s = (-2 \pm j2)$. The damping ratio of the system is

A. 1 **B.** 0.8 **C.** 0.707 **D.** 0.6

Q1.04.2 MCQ 2004 2M

The torque-speed curve of a constant field armature controlled DC servomotor is shown in Fig. The armature resistance in Ω and torque constant in Nm/A of the motor respectively are

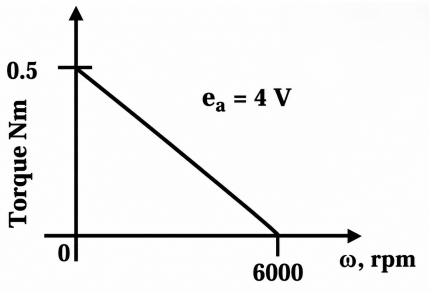

Fig. Q1.04.2

A. (1.76, 0.68) **B.** (1.76, 0.85)

C. (2.00, 0.25) **D.** (0.01, 0.81)

Q1.03.1 MCQ 2003 1M

A three-stack variable reluctance stepper motor has 12 numbers of rotor and stator teeth on each stack. The resolution of angular rotation for the stepper motor, assuming no two stacks are excited simultaneously, is

A. 10° **B.** 15° **C.** 20° **D.** 30°

Q1.03.2 MCQ 2003 1M

In a synchro pair, the control transmitter excites the three stator windings of the control transformer. The stator winding voltages will have

A. equal magnitudes but different phases

B. different magnitudes and different phases

C. equal magnitudes and phases

D. different magnitudes but equal phases

Q1.98.1 MCQ 1998 4M

Match the transfer functions with the systems

(a) All pass filter	(e) e^{-as}
(b) Transport delay	(f) $\dfrac{1-s}{1+s}$
(c) Lag network	(g) $\dfrac{1+as}{1+bs}, a < b$
(d) Servomotor	(h) $\dfrac{K}{s(1+as)}$

Q1.96.1 MCQ 1996 2M

Match the characteristics with the system transfer function.

(a) All pass

(b) Transport delay

(c) Lead compensator

(d) Unstable

(e) $e^{-0.5s}$

(f) $\dfrac{1-s}{1+s}$

(g) $\dfrac{1}{s^2+2s-3}$

(h) $\dfrac{s}{s^2+s+100}$

(i) $\dfrac{10(1+0.3s)}{(1+0.1s)}$

Q1.94.1 MCQ 1994 1M

In a second order instrument, 'K' is the spring constant, 'M' is the mass of the instrument, and 'B' is the damping constant. The damping constant necessary to just prevent overshoot in the step response is

A. equal to $\sqrt{KM}$ **B.** less than $\sqrt{KM}$

C. greater than $\sqrt{KM}$ **D.** equal to $2\sqrt{KM}$

Q1.94.2 MCQ 1994 1M

A first order process has a static gain of 1, a time constant of 1 sec and transport delay of 0.1 sec. Its transfer function is

A. $\dfrac{e^{-s}}{1+0.1s}$ **B.** $\dfrac{e^{-10s}}{1+s}$

C. $\dfrac{e^{-0.1s}}{1+s}$ **D.** $\dfrac{e^{-0.1s}}{1+0.1s}$

Q1.94.3 MCQ 1994 4M

Match the elements and the transfer functions :

(a) pneumatic actuator

(b) a transport delay

(c) a servomotor

(d) a nozzle-flapper

(e) e^{-sT}

(f) K

(g) $\dfrac{K}{as^2+bs+c}$

(h) $\dfrac{K}{s(1+as)}$

Q1.93.1 NAT 1993 2M

A synchro transmitter is used with a synchro repeater for _____ and with a control transformer for _____

Q1.92.1 MCQ 1992 1M

If the open loop transfer function of negative feedback control system is $G(s)H(s) = \dfrac{K}{(s+1)^3}$, then the gain K for a closed loop pole to be at $\left(-\dfrac{1}{2}+j\dfrac{\sqrt{3}}{2}\right)$ is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q1.92.2 NAT 1992 2M

A system has a complex pole pair of $(-1 \pm j2)$ and a real zero of (-3). The steady state output to a unit step input is 2. The transfer function of the system is _____.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q1.24.1 **B**	Q1.22.1 **B**	Q1.19.1 **A**
Q1.17.1 **A**	Q1.17.2 **C**	Q1.16.1 **A**
Q1.10.1 **B**	Q1.07.1 **C**	Q1.06.1 **D**
Q1.05.1 **D**	Q1.05.2 **MTA**	Q1.04.1 **C**
Q1.04.2 **C**	Q1.03.1 **A**	Q1.03.2 **A**
Q1.98.1 **a-f, b-e, c-g, d-h**	Q1.96.1 **a-f, b-e, c-i, d-g**	Q1.94.1 **D**
Q1.94.2 **C**	Q1.94.3 **a-f, b-e, c-h, d-g**	Q1.93.1 **amplification, error detection**
Q1.92.1 **A**	Q1.92.2 *	

UNIT

II

Block Diagram and Signal Flow Graph

Topics

1. Block Diagram Reduction Techniques
2. Signal Flow Graphs and Mason's Gain Formula
3. Limitations of Mason's Gain Formula

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Unit Snapshot*

Stream: IN	**Questions:** 12	**MCQ:** 9	**MSQ:** 0	**NAT:** 3
1M: 1	**2M:** 9	**Descriptive:** 2	**Avg Weightage** ~3M	**Range:** 2M(2022)–2M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q2.22.1 MCQ 2022 2M

The signal flow graph of a system is shown. The expression for $Y(s)/X(s)$ is

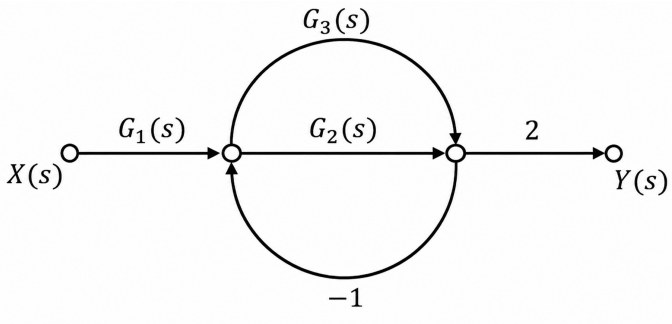


Fig. Q2.22.1

A. $\dfrac{2G_1(s)G_2(s) + 2G_1(s)G_3(s)}{1 + G_2(s) + G_3(s)}$

B. $2 + G_1(s) + G_3(s) + \dfrac{G_2(s)}{1 + G_2(s)}$

C. $G_1(s) + G_3(s) - \dfrac{G_2(s)}{2 + G_2(s)}$

D. $\dfrac{2G_1(s)G_2(s) + 2G_1(s)G_3(s) - G_1(s)}{1 + G_2(s) + G_3(s)}$

Q2.17.1 MCQ 2017 2M

The overall closed loop transfer function $\dfrac{C(s)}{R(s)}$, represented in the figure, will be

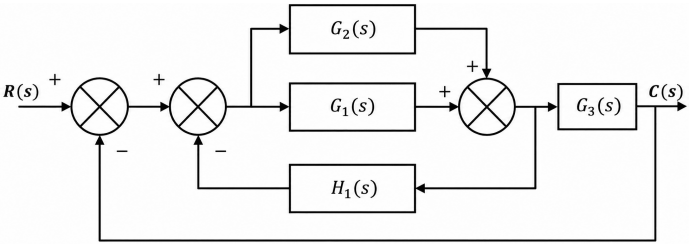


Fig. Q2.17.1

A. $\dfrac{(G_1(s) + G_2(s))G_3(s)}{1 + (G_1(s) + G_2(s))(H_1(s) + G_3(s))}$

B. $\dfrac{(G_1(s) + G_3(s))}{1 + G_1(s)H_1(s) + G_2(s)G_3(s)}$

C. $\dfrac{(G_1(s) - G_2(s))H_1(s)}{1 + (G_1(s) + G_3(s))(H_1(s) + G_1(s))}$

D. $\dfrac{G_1(s)G_2(s)H_1(s)}{1 + G_1(s)H_1(s) + G_1(s)G_3(s)}$

Q2.14.1 MCQ 2014 2M

Consider the control system shown in figurewith feedforwardaction for rejection of a measurable disturbance $d(t)$. The value of K, for whichthe disturbance response at the output $y(t)$ is zero mean, is:

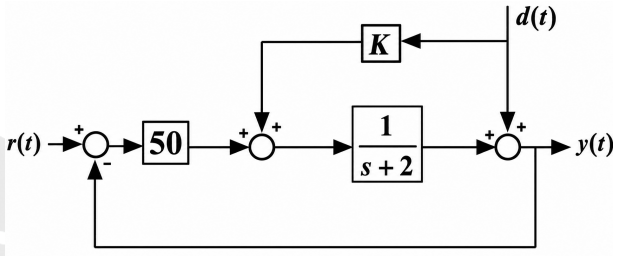


Fig. Q2.14.1

A. 1 **B.** -1 **C.** 2 **D.** -2

Q2.13.1 MCQ 2013 2M

The signal flow graph for a system is given below. The transfer function $\dfrac{Y(s)}{U(s)}$ for this system is given as

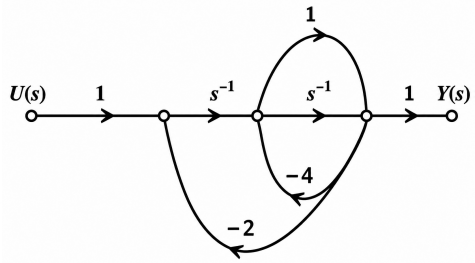


Fig. Q2.13.1

A. $\dfrac{s+1}{5s^2 + 6s + 2}$ **B.** $\dfrac{s+1}{s^2 + 6s + 2}$

C. $\dfrac{s+1}{s^2 + 4s + 2}$ **D.** $\dfrac{1}{5s^2 + 6s + 2}$

Q2.11.1 MCQ 2011 2M

The signal flow graph of a system is given below.

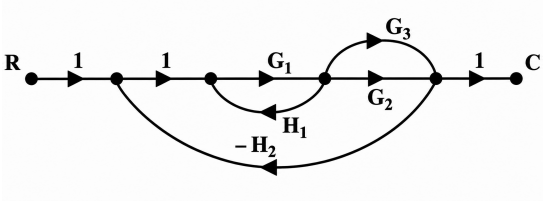


Fig. Q2.11.1

The transfer function (C/R) of the system is

A. $\dfrac{(G_1G_2 + G_1G_3)}{(1 + G_1G_2H_2)}$

B. $\dfrac{(G_1G_2 + G_1G_3)}{(1 - G_1H_1 + G_1G_2H_2)}$

C. $\dfrac{(G_1G_2 + G_1G_3)}{(1 - G_1H_1 + G_1G_2H_2 + G_1G_3H_2)}$

D. $\dfrac{(G_1G_2 + G_1G_3)}{(1 - G_1H_1 + G_1G_2H_2 + G_1G_3H_2 + G_1G_2G_3H_1)}$

Q2.09.1 MCQ 2009 2M

A filter is represented by the signal flow graph shown in the figure. Its input is $x(t)$ and output is $y(t)$. The transfer function of the filter is

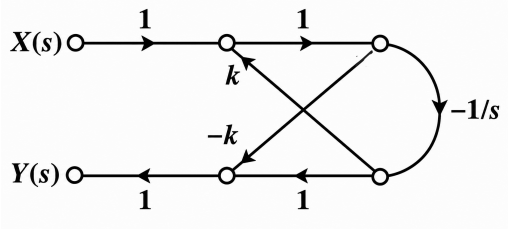


Fig. Q2.09.1

A. $\dfrac{-(1 + ks)}{s + k}$ **B.** $\dfrac{1 + ks}{s + k}$ **C.** $\dfrac{-1 + ks}{s + k}$ **D.** $\dfrac{1 - ks}{s + k}$

Q2.06.1 MCQ 2006 2M

The signal flow graph representation of a control system is shown below.The transfer function $\dfrac{Y(s)}{R(s)}$ is computed as

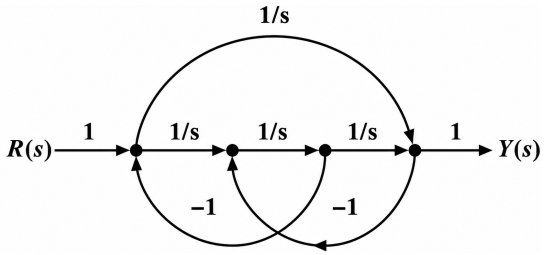


Fig. Q2.06.1

A. $\dfrac{1}{s}$ **B.** $\dfrac{s^2 + 1}{s(s^2 + 2)}$ **C.** $\dfrac{s(s^2 + 1)}{s^2 + 2}$ **D.** $1 - \dfrac{1}{s}$

Q2.05.1 MCQ 2005 2M

The transfer function of the open loop system $G(s)$ which is represented by the signal flow graph shown the figure below is

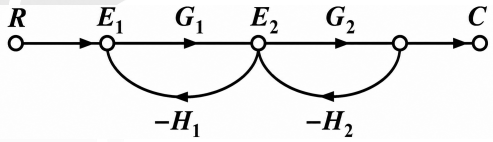


Fig. Q2.05.1

A. $\dfrac{G_1G_2}{1 + G_1H_1 + G_2H_2}$ **B.** $\dfrac{G_1G_2}{1 + G_1H_1G_2H_2}$

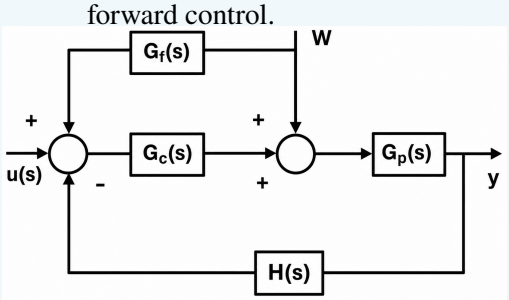
C. $\dfrac{G_1G_2}{(1 + G_1H_1)(1 + G_2H_2)}$ **D.** $\dfrac{G_1G_2}{1 + (G_1 + H_1)(G_2 + H_2)}$

Common Data Questions

The next 2 questions share the data below.

Common Data for Q2.00.1 and Q2.00.2.

Fig. shows the block diagram for a system with feed forward control.



Q2.00.1 NAT 2000 5M

Find the transfer function between the output y and the disturbance input w.

Q2.00.2 NAT 2000 5M

What would be the transfer function of the feed forward controller $G_f(s)$ to eliminate completely the effect of the disturbance input w?

Q2.99.1 MCQ 1999 1M

The transfer function between y_2 and y_1 in Fig. is

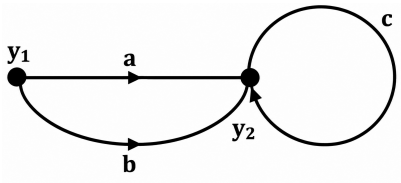


Fig. Q2.99.1

A. $a + b$ **B.** $(a + b)c$ **C.** $\frac{a+b}{1-c}$ **D.** $\frac{a+b}{1+c}$

Q2.95.1 NAT 1995 2M

For the signal flow graph show in figure, the transfer function is $C(s)/R(s) =$ _____.

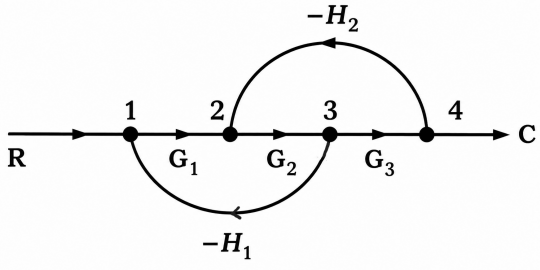


Fig. Q2.95.1

Q2.22.1 A	Q2.17.1 A	Q2.14.1 D
Q2.13.1 A	Q2.11.1 C	Q2.09.1 A
Q2.06.1 A	Q2.05.1 A	Q2.00.1 *
Q2.00.2 *	Q2.99.1 C	Q2.95.1 *

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

UNIT

III

Time Response Analysis

Topics

1. Impulse, Step, Ramp and Parabolic Response
2. Concept of Damping in 2nd Order Systems
3. Complex Frequency and Damping in RLC Circuits
4. Step Response of 2nd Order Systems
5. Time Domain Performance Parameters
6. Steady State Error — Unity and Non-Unity Feedback
7. Error Constants and System Types

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 40	**MCQ:** 28	**MSQ:** 1	**NAT:** 11
1M: 13	**2M:** 26	**Descriptive:** 1	**Avg Weightage** ~3M	**Range:** 1M(2014)–4M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 NAT 2026 2M

A linear time invariant (LTI) system has a transfer function

$$G(s) = \frac{10(s+1)}{s(s^2+2s+5)}$$

The system is placed in unity negative feedback configuration. For a unit ramp reference, the steady state error is *(rounded off to two decimal places)*

Q3.25.1 NAT 2025 2M

A feedback control system is shown in the figure.

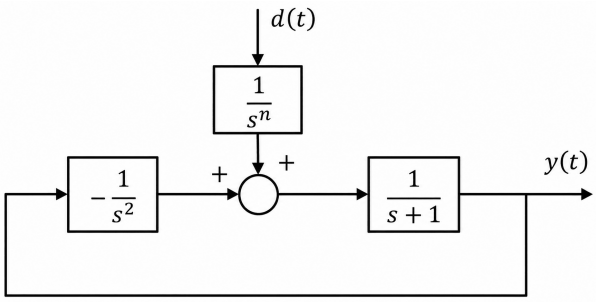


Fig. Q3.25.1

The maximum allowable value of n such that the output $y(t)$, due to any step disturbance signal $d(t)$, becomes zero at steady-state, is

---

Q3.24.1 MCQ 2024 2M

Consider the feedback control system shown in the figure. The steady-state error $e_{ss} = \lim_{t\to\infty}(r(t) - y(t))$ due to unit step reference $r(t)$ is

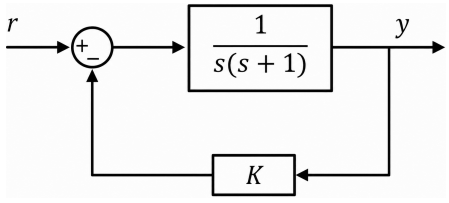


Fig. Q3.24.1

A. $\frac{K-1}{K}$ **B.** $\frac{1}{2}$ **C.** 0 **D.** $\frac{1-K}{K}$

Q3.24.2 MCQ 2024 2M

The transfer function of a system is

$$G(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

Choose the range of ξ and ω_n (in rad/s) from the following options such that the poles lie on the shaded region of the s-plane as shown in the figure.

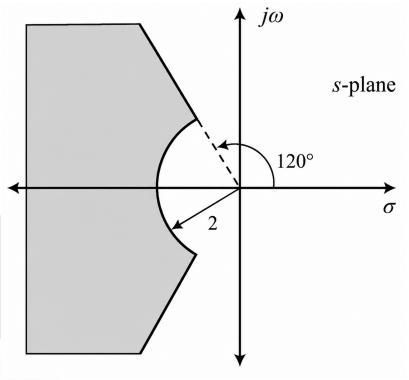


Fig. Q3.24.2

A. $\xi \geq \frac{1}{2}$ and $\omega_n \geq 2$ **B.** $\xi \geq \frac{1}{4}$ and $\omega_n \geq 2$

C. $\xi \geq \frac{1}{2}$ and $\omega_n \geq \sqrt{3}$ **D.** $\xi \geq \frac{1}{4}$ and $\omega_n \geq \sqrt{3}$

Q3.21.1 MSQ 2021 1M

The step response of a circuit is seen to have an oscillatory behaviour at the output with oscillations dying down after some time. The correct inference(s) regarding the transfer function from input to output is/are

- ☐ **A.** that it is of at least second order.
- ☐ **B.** that it has at least one pole-pair that is under-damped.
- ☐ **C.** that it does not have a real pole.
- ☐ **D.** that it is a first order system.

Q3.21.2 NAT 2021 2M

Consider a system with transfer-function $G(s) = \frac{2}{s+1}$. A unit step function $\mu(t)$ is applied to the system, which results in an output $y(t)$. If $e(t) = y(t) - \mu(t)$, then $\lim_{t\to\infty} e(t)$ is

---

Q3.20.1 NAT 2020 1M

A second order system has closed loop poles located at $s = -3 \pm j4$. The time t at which the maximum value of the step response occurs (in seconds, rounded off to two decimal places) is _____

Q3.20.2 NAT 2020 2M

The system shown in Fig. (a) has a time response $y(t)$ to an input $r(t) = 10\,u(t)$ as shown in Fig. (b), $u(t)$ being the unit step input. Both K, τ are positive. The gain K of the system is _____

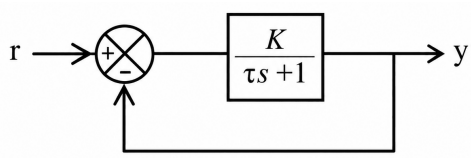


Fig. (a)

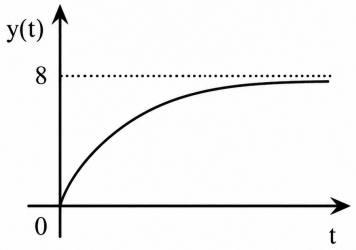


Fig. (b)

Fig. Q3.20.2

Q3.19.1 NAT 2019 2M

The transfer function relating the input $x(t)$ to the output $y(t)$ of a system is given by $G(s) = \dfrac{1}{(s+3)}$. A unit-step input is applied to the system at time $t = 0$. Assuming that $y(0) = 3$, the value of $y(t)$ at time $t = 1$ is (Answer should be rounded off to two decimal places)

_____

Q3.17.1 NAT 2017 2M

The block diagram of a closed-loop control system is shown in the figure. The values of k and k_p are such that the system has a damping ratio of 0.8 and an undamped natural frequency ω_n of 4 rad/s respectively. The value of k_p will be _____ (rounded off to two decimal places).

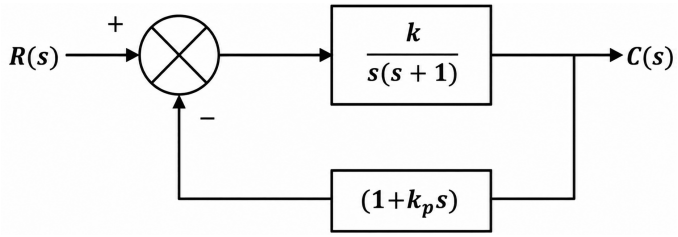


Fig. Q3.17.1

Q3.16.1 MCQ 2016 2M

Match the unit-step responses (1), (2) and (3) with the transfer functions $P(s)$, $Q(s)$ and $R(s)$, given below.

$$P(s) = \frac{-1}{(s+1)};\ Q(s) = \frac{2(s-1)}{(s+10)(s+2)};\ R(s) = \frac{1}{(s+1)^2}$$

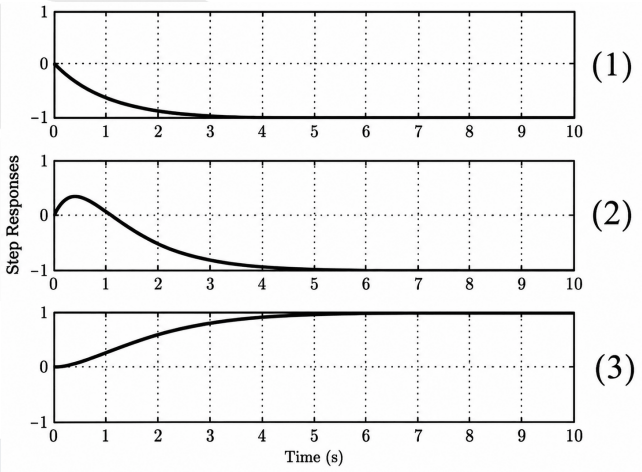


Fig. Q3.16.1

A. $P(s) - (3), Q(s) - (2), R(s) - (1)$

B. $P(s) - (1), Q(s) - (2), R(s) - (3)$

C. $P(s) - (2), Q(s) - (1), R(s) - (3)$

D. $P(s) - (1), Q(s) - (3), R(s) - (2)$

Q3.15.1 MCQ 2015 1M

A mass-spring-damper system with force as input and displacement of the mass as output has a transfer function $G(s) = \dfrac{1}{(s^2 + 24s + 900)}$. A force input $F(t) = 10\sin(70t)$ newtons is applied at time $t = 0$ s. A beam from an optical stroboscope is focused on the mass. In steady state, the strobe frequency in hertz at which the mass appears to be stationary is

A. $\frac{5}{\pi}$ **B.** $\frac{15}{\pi}$ **C.** $\frac{35}{\pi}$ **D.** $\frac{50}{\pi}$

Q3.15.2 NAT 2015 2M

A system with transfer function $G(s) = \frac{1}{s^2+1}$ has zero initial conditions. The percentage overshoot in its step response is (in %)

Q3.14.1 MCQ 2014 1M

A plant has an open-loop transfer function,

$$G_p(s) = \frac{20}{(s+0.1)(s+2)(s+100)}.$$

The approximate model obtained by retaining only one of the above poles, which is closest to the frequency response of the original transfer function at low frequency is

A. $\frac{0.1}{s+0.1}$ **B.** $\frac{2}{s+2}$ **C.** $\frac{100}{s+100}$ **D.** $\frac{20}{s+0.1}$

Q3.13.1 MCQ 2013 2M

The open-loop transfer function of a dc motor is given as $\frac{\omega(s)}{V_a(s)} = \frac{10}{1+10s}$. When connected in feedback as shown below, the approximate value of K_a that will reduce the time constant of the closed loop system by one hundred times as compared to that of the open-loop system is

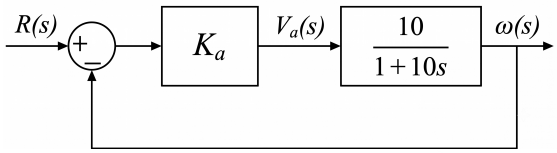


Fig. Q3.13.1

A. 1 **B.** 5 **C.** 10 **D.** 100

Q3.11.1 MCQ 2011 2M

The unit-step response of a negative unity feedback system with the open-loop transfer function $G(s) = \frac{6}{s+5}$ is

A. $1 - e^{-5t}$ **B.** $6 - 6e^{-5t}$

C. $\frac{6}{5} - \frac{6}{5}e^{-5t}$ **D.** $\frac{6}{11} - \frac{6}{11}e^{-11t}$

Common Data Questions

Data for Q3.11.2.

The open-loop transfer function of a unity negative feedback control system is given by

$$G(s) = \frac{K}{(s+5)^3}$$

The 1st Question is given in Unit VI (Q6.11.1)

Q3.11.2 MCQ 2011 2M

The value of K for the damping ratio ζ to be 0.5, corresponding to the dominant closed-loop complex conjugate pole pair is

A. 250 **B.** 125 **C.** 75 **D.** 50

Q3.10.1 MCQ 2010 2M

A unit ramp input is applied to the system shown in the adjoining figure. The steady state error in its output is

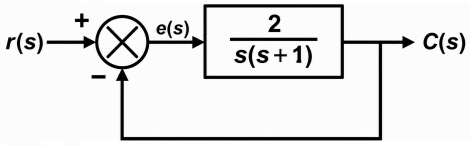


Fig. Q3.10.1

A. 0 **B.** 0.5 **C.** 1 **D.** 2

Q3.10.2 MCQ 2010 2M

A unity feedback system has an open loop transfer function $G(s) = \frac{k}{s(s+3)}$. The value of k that yields a damping ratio of 0.5 for the closed loop system is

A. 1 **B.** 3 **C.** 5 **D.** 9

Q3.09.1 MCQ 2009 2M

A unity feedback system has the transfer function $\frac{K(s+b)}{s^2(s+20)}$ The value of b for which the loci of all the three roots of the closed loop characteristic equation meet at a single point is

A. $\frac{10}{9}$ **B.** $\frac{20}{9}$ **C.** $\frac{30}{9}$ **D.** $\frac{40}{9}$

Common Data Questions

The next 2 questions share the data below.

Common data for Q3.08.1 and Q3.08.2

A unity feedback system has open loop transfer function $G(s) = \dfrac{100}{s(s+p)}$. The time at which the response to a unit step input reaches its peak is $\dfrac{\pi}{8}$ seconds.

Q3.08.1 MCQ 2008 2M

The damping coefficient for the closed loop system is

A. 0.4 **B.** 0.6 **C.** 0.8 **D.** 1

Q3.08.2 MCQ 2008 2M

The value of p is

A. 6 **B.** 12 **C.** 14 **D.** 16

Q3.05.1 MCQ 2005 2M

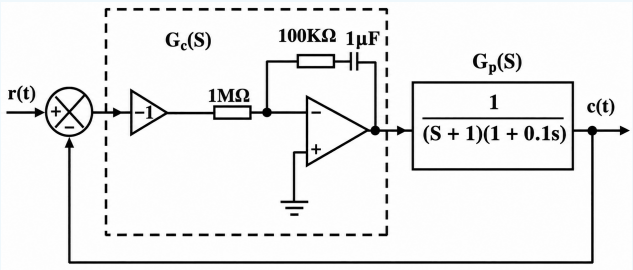
Consider a unity feedback control system whose forward path transfer function $G(s) = K/s^2$. The steady state error, for a step input is

A. 1 **B.** ∞ **C.** 0 **D.** does not exist

Common Data Questions

The next 2 questions share the data below.

Common data for Q3.05.2 and Q3.05.3



Given figure shows a closed loop unity feedback system. The controller block has a transfer function denoted by $G_c(s)$. The controller is cascaded to the plant, which is denoted by $G_p(s)$.

Q3.05.2 MCQ 2005 2M

The transfer function $G_c(s)$ is

A. $\dfrac{1+0.1s}{s}$ **B.** $-\dfrac{1+0.1s}{s}$ **C.** $-\dfrac{s}{s+1}$ **D.** $\dfrac{s}{s+1}$

Q3.05.3 MCQ 2005 2M

The peak percentage overshoot of the closed loop system is

A. 5.0% **B.** 10.0% **C.** 16.3% **D.** 1.63%

Q3.04.1 MCQ 2004 1M

The steady state error due to a unit-step disturbance input $D(s)$ in figure is

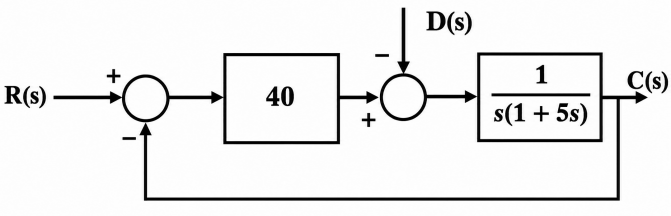


Fig. Q3.04.1

A. 0 **B.** 0.012 **C.** 0.021 **D.** 0.025

Q3.04.2 MCQ 2004 1M

For a first order instrument a 5% settling time is equal to

A. three times the time constant

B. two times the time constant

C. the time constant

D. time required for the output signal to reach 5% of the final value

Q3.04.3 MCQ 2004 2M

A certain system exhibited an overshoot of 16% when subjected to input of $2u(t)$, where $u(t)$ is step input. The damping ratio and decay ratio respectively are

A. (0.8, 0.0810) **B.** (0.5, 0.0256)
C. (1.0, 0.1626) **D.** (1.1, 0.0089)

Q3.03.1 MCQ 2003 1M

In the feedback scheme shown in Fig., the time-constant of the closed loop system will be

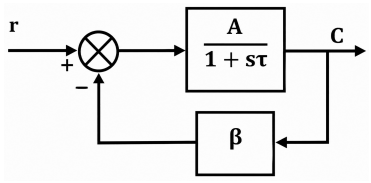


Fig. Q3.03.1

A. $A\beta\tau$ **B.** τ **C.** $(1 + A\beta, \tau)$ **D.** $\dfrac{\tau}{(1 + A\beta)}$

Q3.03.2 MCQ 2003 1M

The steady state error for the feedback control system shown in Fig. when subjected to a unit step input is

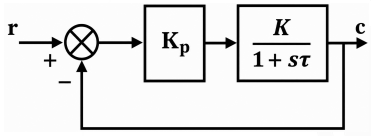


Fig. Q3.03.2

A. 0 **B.** $\dfrac{1}{(1 + KK_P)}$ **C.** $\dfrac{KK_p}{(1 + KK_P)}$ **D.** ∞

Q3.02.1 MCQ 2002 1M

A unity feed back system has a forward path transfer function $G(s) = \dfrac{10}{s(s + 5)}$. It is subjected to a unity ramp input. The integral of the squared error (ISE) is

A. 0 **B.** ∞ **C.** always negative **D.** 0.5

Q3.02.2 MCQ 2002 2M

The forward path transfer function of a unity feedback system is given by $G(s) = \dfrac{(1 + 5s)(1 + 10s)(1 + 2s)}{(1 + s)(1 + 8s)(1 + 20s)}$.

If $e(t)$ is the error to a unit impulse input, the value of performance index $J = \int_0^\infty e(t)\, dt$ is

A. zero **B.** ∞ **C.** -12 **D.** 0.5

Q3.02.3 MCQ 2002 2M

The loop transfer function of a system is given by $G(s)H(s) = \dfrac{10e^{-0.1s}}{s(s + 5)(s^2 + 4s + 16)}$. When the system is subjected to a parabolic input of $t^2/2$, the steady state error is

A. zero **B.** 0.1 **C.** ∞ **D.** $e^{-0.1}/0.1$

Q3.02.4 NAT 2002 5M

The block diagram of a second order servomechanism is shown in Fig. Tests conducted on the servomechanism resulted in the following:

(i) A unit step response produced a maximum overshoot of 16%.

(ii) Keeping the reference shaft fixed, a torque of 10 Nm applied to the output shaft resulted in a steady state error of 0.2 rad.

Find the parameters of the servo-mechanism. J and K. (assume the gear ratio $n = 0.1$ and $B = 0.125$ Nm/rad/s.)

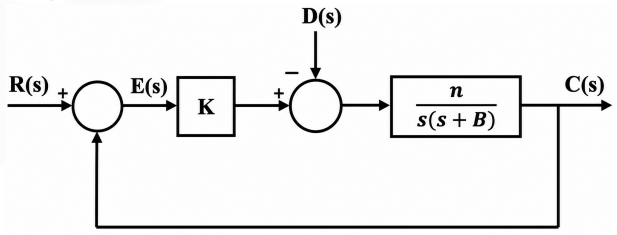


Fig. Q3.02.4

Q3.00.1 MCQ 2000 1M

For a unity feedback system with open loop transfer function $G(s) = \dfrac{9}{s(s + 2)}$, the damping ratio is

A. 1/3 **B.** 1/2 **C.** 1 **D.** 2

Q3.99.1 MCQ 1999 1M

A system with transfer function $\dfrac{1}{\tau s + 1}$, subjected to a step input takes 10 seconds to reach 50% of the step height. The value of τ is

A. 6.9 s **B.** 10 s **C.** 14.4 s **D.** 20 s

Q3.97.1 MCQ 1997 1M

The transfer function of a control system is given by: $\frac{C(s)}{R(s)} = \frac{25}{s^2 + 6s + 25}$. The first maximum value of the response occurs at a time t_{max} given by

A. $\frac{\pi}{8}$ **B.** $\frac{\pi}{4}$ **C.** $\frac{\pi}{2}$ **D.** π

Q3.96.1 NAT 1996 2M

Match the roots in the s-plane with the impulse response shown in figure

Column I : Impulse Response

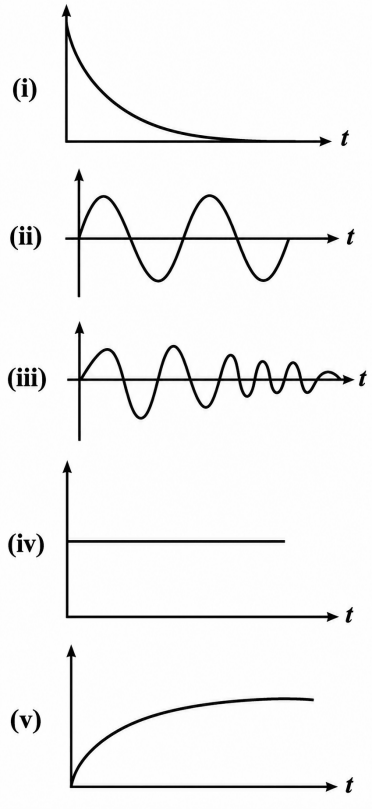


Fig. Q3.96.1

Column II : Roots in the s-plane

A. Two imaginary Roots

B. Two complex roots in the left half plane

C. A single root on the negative real axis

D. A single root at the origin

Q3.94.1 MCQ 1994 1M

A unity feedback closed loop second order system has a transfer function $\frac{8}{s^2 + 0.6s + 9}$ and is excited by a step input of 1 unit. The steady state error of the output is

A. 10 **B.** 0.0 **C.** 1.0 **D.** 0.1

Q3.93.1 NAT 1993 2M

A unit step is applied at $t = 0$ to a first order system without time delay. The response has a value of 1.264 units at $t = 10$ mins. and 2 units at steady state. The transfer function of the system is

_____

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.
MTA Marks To All awarded — see detailed solutions.

Q3.26.1 0.49 to 0.51	Q3.25.1 MTA	Q3.24.1 A
Q3.24.2 A	Q3.21.1 A,B	Q3.21.2 1
Q3.20.1 0.78 to 0.79	Q3.20.2 4	Q3.19.1 0.46 to 0.48
Q3.17.1 0.32 to 0.4	Q3.16.1 MTA	Q3.15.1 C
Q3.15.2 100	Q3.14.1 A	Q3.13.1 C
Q3.11.1 D	Q3.11.2 B	Q3.10.1 B
Q3.10.2 D	Q3.09.1 B	Q3.08.1 B
Q3.08.2 B	Q3.05.1 C	Q3.05.2 A
Q3.05.3 C	Q3.04.1 D	Q3.04.2 A
Q3.04.3 B	Q3.03.1 D	Q3.03.2 B
Q3.02.1 B	Q3.02.2 D	Q3.02.3 C
Q3.02.4 50, 3.08×10^{-3}	Q3.00.1 A	Q3.99.1 C

Q3.97.1 B	Q3.96.1 A-(ii), B-(iii), C-(i), D-(iv)	Q3.94.1 D
Q3.93.1 *		

PrepFusion

UNIT

IV

Routh's Stability Criterion

Topics

1. Hurwitz Polynomial and Routh Array
2. Special Cases in Routh Stability
3. Mapping Theorem in Routh-Hurwitz

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 14	**MCQ:** 11	**MSQ:** 0	**NAT:** 3
1M: 5	**2M:** 9	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–3M(2016)

**See preface for how Avg Weightage and Range are calculated.*

Q4.25.1 MCQ 2025 1M

For the transfer function $G(s) = 1 + \dfrac{2s-1}{s^3+5s^2+3s+22}$, the number of zeros lying in the left half of the s-plane is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q4.23.1 NAT 2023 1M

The number of zeros of the polynomial $P(s) = s^3 + 2s^2 + 5s + 80$ in the right-half plane is

_____

Q4.18.1 NAT 2018 2M

Consider a standard negative feedback configuration with $G(s) = \dfrac{1}{(s+1)(s+2)}$ and $H(s) = \dfrac{s+\alpha}{s}$. For the closed loop system to have poles on the imaginary axis, the value of α should be equal to (up to one decimal place)

_____

Q4.17.1 MCQ 2017 2M

A closed-loop system is shown in the figure. The system parameter α is not known. The condition for asymptotic stability of the closed loop system is

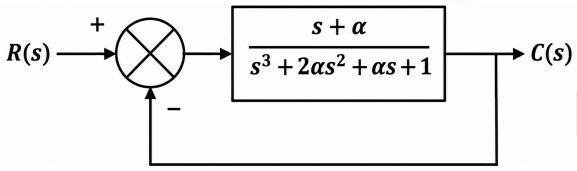


Fig. Q4.17.1

A. $\alpha < -0.5$ **B.** $-0.5 < \alpha < 0.5$
C. $0 < \alpha < 0.5$ **D.** $\alpha > 0.5$

Q4.16.1 MCQ 2016 1M

The value of a_0 which will ensure that the polynomial $s^3 + 3s^2 + 2s + a_0$ has roots on the left half of the s-plane is

A. 11 **B.** 9 **C.** 7 **D.** 5

Q4.16.2 MCQ 2016 2M

For the feedback system given below, the transfer function $G(s) = \dfrac{1}{(s+1)^2}$. The system **CANNOT** be stabilized with

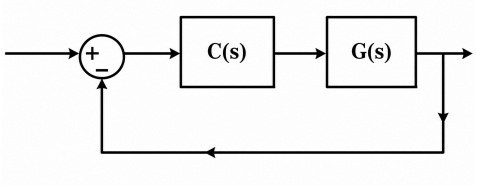


Fig. Q4.16.2

A. $C(s) = 1 + \dfrac{3}{s}$ **B.** $C(s) = 3 + \dfrac{7}{s}$
C. $C(s) = 3 + \dfrac{9}{s}$ **D.** $C(s) = \dfrac{1}{s}$

Q4.12.1 MCQ 2012 2M

The feedback system shown below oscillates at 2 rad/s when

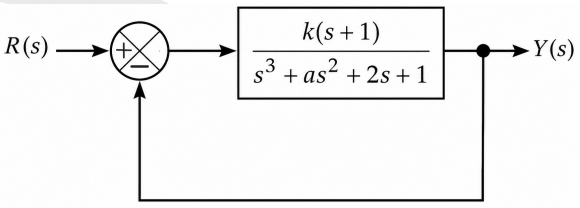


Fig. Q4.12.1

A. $K = 2$ and $a = 0.75$ **B.** $K = 3$ and $a = 0.75$
C. $K = 4$ and $a = 0.5$ **D.** $K = 2$ and $a = 0.5$

Q4.11.1 MCQ 2011 1M

The first two rows of Routh's table of a third-order characteristic equation are

s^3	3	3
s^2	4	4

It can be inferred that the system has

A. one real pole in the right-half of s-plane

B. a pair of complex conjugate poles in the right-half of s-plane

C. a pair of real poles symmetrically placed around $s = 0$

D. a pair of complex conjugate poles on the imaginary axis of the s-plane

Q4.10.1 MCQ 2010 1M

The open loop transfer function of a unity gain feedback system is given by:

$$G(s) = \frac{k(s+3)}{(s+1)(s+2)}.$$

The range of positive values of k for which the closed loop system will remain stable is:

A. $1 < k < 3$ **B.** $0 < k < 10$
C. $5 < k < \infty$ **D.** $0 < k < \infty$

Q4.08.1 MCQ 2008 2M

A closed loop control system is shown below. The range of the controller gain K_C which will make the real parts of all the closed loop poles more negative than -1 is

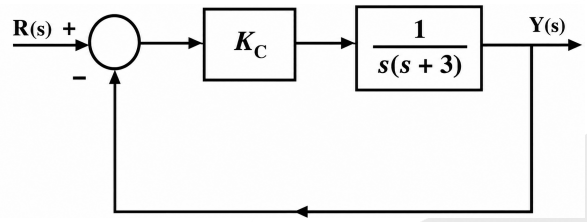


Fig. Q4.08.1

A. $K_C > -4$ **B.** $K_C > 0$ **C.** $K_C > 2$ **D.** $K_C < 2$

Q4.04.1 MCQ 2004 2M

A certain closed loop system with unity feedback has the forward path transfer function given by

$$G(s) = \frac{K}{s(s+1)(s+5)}$$

With the gain set at the ultimate value, the system will oscillate at an angular frequency of

A. $\sqrt{6}$ rad/sec **B.** $\sqrt{5}$ rad/sec
C. $\sqrt{2}$ rad/sec **D.** $\sqrt{3}$ rad/sec

Q4.03.1 MCQ 2003 2M

The closed loop control system shown in figure has $\tau_1 > 0$. The system will remain stable for all τ_1 in the range

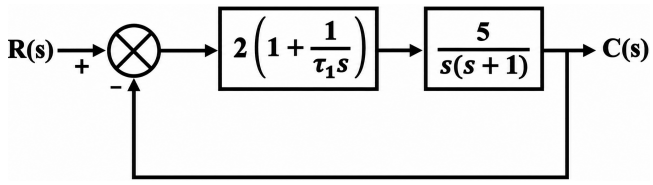


Fig. Q4.03.1

A. $\tau_1 < 0.5$ **B.** $0.5 < \tau_1 < 1.0$
C. $\tau_1 < 1.0$ **D.** $\tau_1 > 1.0$

Q4.02.1 MCQ 2002 2M

The partial Routh array of the characteristic equation of a system is given by;

s^4	1	a	8
s^3	3	12	

The system oscillates with a frequency of 2 rad/s. The value of the parameter 'a' of the system is

A. 6 **B.** 2 **C.** 8 **D.** 12

Q4.93.1 NAT 1993 2M

The characteristic equation of a closed loop system is $s^2 + Ks + 4K - 1 = 0$. The system is stable for K greater than _____

Answer Key

Q4.25.1 D	Q4.23.1 2	Q4.18.1 8.9 to 9.1
Q4.17.1 D	Q4.16.1 D	Q4.16.2 C
Q4.12.1 A	Q4.11.1 D	Q4.10.1 D
Q4.08.1 C	Q4.04.1 B	Q4.03.1 D
Q4.02.1 A	Q4.93.1 0.25	

UNIT

V

Root Locus

Topics

1. Basics of Root Locus
2. Rules for Sketching Root Locus
3. Angle of Departure and Arrival
4. Complementary Root Locus

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 17	**MCQ:** 12	**MSQ:** 0	**NAT:** 5
1M: 4	**2M:** 13	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2014)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q5.24.1 NAT 2024 2M

Consider a unity negative feedback system with its open-loop pole-zero map as shown in the figure. If the point $s = j\alpha$, $\alpha > 0$, lies on the root locus, the value of α is

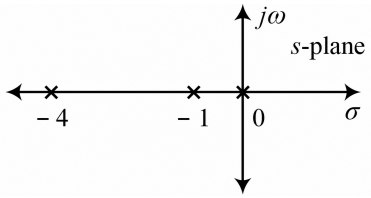


Fig. Q5.24.1

Q5.20.1 NAT 2020 2M

The loop transfer function of a negative feedback system is given by

$$G(s)H(s) = \frac{K}{s(s+2)(s+6)}, \text{ where } K > 0.$$

The value of K at the breakaway point of the root locus for the above system (rounded off to one decimal place) is

Q5.19.1 MCQ 2019 2M

In a control system with unity gain feedback, the plant has the transfer function $P(s) = \frac{3}{s}$. Assuming that a controller of the form $C(s) = \frac{K}{(s+p)}$ is used, where K is a positive constant, the value of p for which the root-locus of the closed-loop system passes through the points $-3 \pm j3\sqrt{3}$ where $j = \sqrt{-1}$, is

A. 3 **B.** $3\sqrt{3}$ **C.** 6 **D.** 9

Q5.17.1 NAT 2017 2M

The loop transfer function of a closed-loop system is given by $G(s)H(s) = \frac{K(s+6)}{s(s+2)}$. The breakaway point of the root-loci will be

Q5.15.1 MCQ 2015 2M

The open loop transfer function of a system is $G(s) = \frac{s^2+6s+10}{s^2+2s+2}$. The angles of arrival of its root loci are

A. $\pm\frac{\pi}{4}$ **B.** $\pm\frac{\pi}{3}$ **C.** $\pm\frac{\pi}{2}$ **D.** $\pm\frac{\pi}{6}$

Q5.14.1 NAT 2014 1M

A loop transfer function is given by :

$$G(s)H(s) = \frac{K(s+2)}{s^2(s+10)}$$

The point of intersection of the asymptotes of $G(s)H(s)$ on the real axis in the s-plane is at

Q5.12.1 MCQ 2012 2M

The open loop transfer function of a unity gain negative feedback control system is given by $G(s) = \frac{s^2+4s+8}{s(s+2)(s+8)}$. The angle θ, at which the root locus approaches the zeros of the system, satisfies

A. $|\theta| = \pi - \tan^{-1}\left(\frac{1}{4}\right)$ **B.** $|\theta| = \frac{3\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)$

C. $|\theta| = \frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right)$ **D.** $|\theta| = \frac{\pi}{4} - \tan^{-1}\left(\frac{1}{3}\right)$

Q5.11.1 MCQ 2011 2M

Consider the second-order system with the characteristic equation $s(s+3) + K(s+5) = 0$. Based on the properties of the root loci, it can be shown that the complex portion of the root loci of the given system for $0 < K < \infty$ is described by a circle, and the two breakaway points on the real axis are

A. $-5 \pm \sqrt{5}/2$ **B.** $-5 \pm \sqrt{5}$

C. $-5 \pm \sqrt{10}$ **D.** $-5 \pm 2\sqrt{5}$

Q5.08.1 MCQ 2008 2M

The open loop transfer function of a unity feedback system is $G(s) = \frac{K(s+2)}{(s+1+j1)(s+1-j1)}$. The root locus plot of the system has

A. two breakaway points located at $s = -0.59$ and $s = -3.41$

B. one breakaway point located at $s = -0.59$

C. one breakaway point located at $s = -3.41$

D. one breakaway point located at $s = -1.41$

Common Data Questions

The next 2 questions share the data below.

Common data for Q5.07.1 and Q5.07.2

A transfer function with unity DC gain has three poles at -1, -2, and -3 and no finite zeros. A plant with this transfer function is connected with a proportional controller of gain K in the forward path, in a unity feedback configuration.

Q5.07.1 MCQ 2007 2M

The transfer function is

A. $\frac{s}{(s-1)(s-2)(s-3)}$ **B.** $\frac{6}{(s+1)(s+2)(s+3)}$

C. $\frac{s}{(s+1)(s+2)(s+3)}$ **D.** $\frac{6}{(s-1)(s-2)(s-3)}$

Q5.07.2 MCQ 2007 2M

If the root locus plot of the closed loop system passes through the points $\pm j\sqrt{11}$, the maximum value of K for stability of the unity feedback closed loop system is

A. $\sqrt{11}$ **B.** 6 **C.** 10 **D.** $6\sqrt{11}$

Q5.06.1 MCQ 2006 2M

The root locus of a plant is given in the following figure. The root locus crosses imaginary axis at $\omega = 4\sqrt{2}$ rad/s with gain $K = 384$. It is observed that the point $s = -1.5 + j1.5$ lies in the root locus. The gain K at $s = -1.5 + j1.5$ is computed as

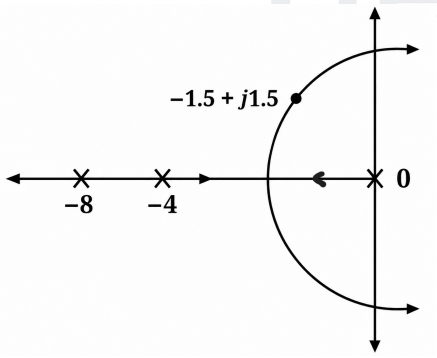


Fig. Q5.06.1

A. 11.3 **B.** 21.2 **C.** 41.25 **D.** 61.2

Q5.05.1 MCQ 2005 2M

The Routh-Hurwitz array is given for a third order characteristic equation.

s^3	1	b	0	0
s^2	a	c	0	0
s^1	$\frac{(6-k)}{3}$	0		
s^0	k			

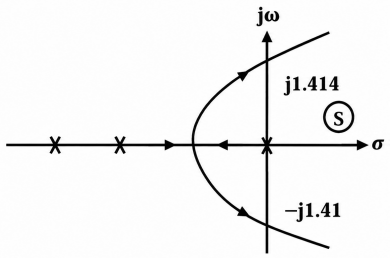


Fig. Q5.05.1

The coefficients a, b, c, and k are such that $a = 3$, $b > 0$, $k > 0$, $c > 0$ and c is a function of k. The root locus for the corresponding characteristic equation is as shown in the given figure. The values of k and c for critical stability respectively are

A. 6 and 6 **B.** 6 and 2

C. 3 and 2 **D.** 3 and 3

Q5.99.1 MCQ 1999 1M

A transfer function has two zeroes at infinity. Then the relation between the numerator degree (N) and the denominator degree (M) of the transfer function is

A. $N = M + 2$ **B.** $N = M - 2$

C. $N = M + 1$ **D.** $N = M - 1$

Q5.95.1 MCQ 1995 1M

The break away point of the root locus on the real axis for a closed-loop system with a loop gain $G(s)H(s) = \frac{K(s+10)}{(s+2)(s+5)}$ lies

A. between −2 and origin **B.** between −2 and −5

C. between −10 and −∞ **D.** at −∞

Q5.94.1 MCQ 1994 1M

The root locus plot of the a unity feedback system is shown in figure. The system transfer function has the form

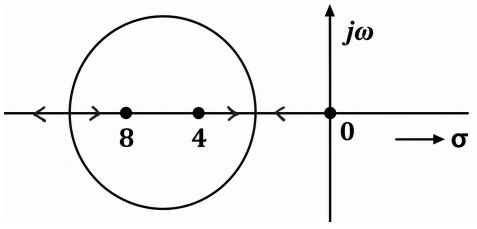


Fig. Q5.94.1

A. $\frac{K}{s(s+4)(s+8)}$ **B.** $\frac{K(s+8)}{s(s+4)}$

C. $\frac{K(s+4)}{s(s+8)}$ **D.** $\frac{Ks}{(s+4)(s+8)}$

Q5.92.1 NAT 1992 2M

A unity feedback system has open loop poles $(-2 \pm j2)$ and 0. It has a single zero at $(-4 \pm j0)$. The angle of departure of the root locus branch starting from the pole $(-2 - j2)$ is

_____

Answer Key

Q5.24.1 2	Q5.20.1 5.0 to 5.1	Q5.19.1 C
Q5.17.1 -1.2 to -1.0	Q5.15.1 A	Q5.14.1 -4
Q5.12.1 D	Q5.11.1 C	Q5.08.1 C
Q5.07.1 B	Q5.07.2 C	Q5.06.1 C
Q5.05.1 A	Q5.99.1 B	Q5.95.1 B
Q5.94.1 B	Q5.92.1 0°	

PrepFusion

UNIT

VI

Polar Plot and Frequency Response of Second Order Systems

Topics

1. Basics of Frequency Response Analysis
2. Frequency Response of 2nd Order Systems
3. Polar Plot
4. Gain and Phase Crossover Frequency
5. Gain Margin and Phase Margin

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 28	**MCQ:** 18	**MSQ:** 0	**NAT:** 10
1M: 10	**2M:** 18	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–4M(2014)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 NAT 2026 2M

The system $G(s) = \dfrac{K}{(s+1)^4}$ has a gain margin of 20 dB. The value of K is _____. *(rounded off to two decimal places)*

Q6.24.1 MCQ 2024 1M

The sinusoidal transfer function corresponding to the polar plot shown in the figure, for $T > 0$, is

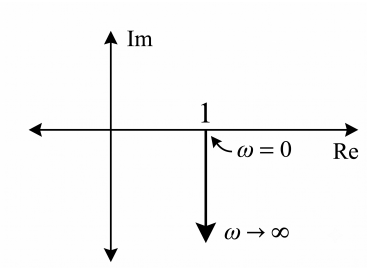


Fig. Q6.24.1

A. $1 - j\omega T$ **B.** $\dfrac{1 - j\omega T}{1 + j\omega T}$ **C.** $1 + j\omega T$ **D.** $\dfrac{1}{1 + j\omega T}$

Q6.23.1 NAT 2023 2M

The phase margin of the transfer function $G(s) = \dfrac{2(1-s)}{(1+s)^2}$ is _____ degrees. **(rounded off to the nearest integer).**

Q6.21.1 NAT 2021 2M

A sinusoid $(\sqrt{2}\sin t)\,\mu(t)$, where $\mu(t)$ is the step input, is applied to a system with transfer-function $G(s) = \dfrac{1}{s+1}$. The amplitude of the steady state output is _____

Q6.20.1 MCQ 2020 1M

The closed-loop transfer function of a control system is given by $\dfrac{C(s)}{R(s)} = \dfrac{1}{1+s}$. For the input $r(t) = \sin t$, the steady-state response $c(t)$ is

A. 1 **B.** $\dfrac{1}{\sqrt{2}}\cos t$

C. $\dfrac{1}{\sqrt{2}}\sin\left(t + \dfrac{\pi}{4}\right)$ **D.** $\dfrac{1}{\sqrt{2}}\sin\left(t - \dfrac{\pi}{4}\right)$

Q6.20.2 NAT 2020 2M

The loop transfer function of a negative feedback system is $G(s)H(s) = \dfrac{2(s+1)}{s^2}$. The phase margin of the system (in degrees, rounded off to one decimal place) is _____

Q6.19.1 MCQ 2019 1M

The loop-gain function $L(s)$ of a control system with unity feedback is given to be

$$L(s) = \frac{k}{(s+1)(s+2)(s+3)},$$

where $k > 0$. If the gain cross-over frequency of the loop-gain function is less than its phase cross-over frequency, the closed-loop system is

A. unstable **B.** marginally stable
C. conditionally stable **D.** stable

Q6.19.2 NAT 2019 2M

In a control system with unity gain feedback, the transfer function of the loop-gain function is $L(s) = \dfrac{9e^{-0.1s}}{s}$. The phase margin of the loop-gain function $L(s)$ is _____ degree.

Q6.18.1 NAT 2018 1M

An input $p(t) = \sin(t)$ is applied to the system $G(s) = \dfrac{s-1}{s+1}$. The corresponding steady state output is $y(t) = \sin(t + \phi)$, where the phase ϕ (in degrees), when restricted to $0° \le \phi \le 360°$, is _____.

Q6.18.2 NAT 2018 1M

Consider the transfer function

$$G(s) = \frac{2}{(s+1)(s+2)}$$

The phase margin of $G(s)$ in degrees is _____.

Q6.16.1 NAT 2016 1M

The input $i(t) = 2\sin(3t + \pi)$ is applied to a system whose transfer function is $G(s) = \dfrac{8}{(s+10)^2}$. The amplitude of the output of the system is

_____.

Q6.14.1 NAT 2014 2M

The transfer function of a system is given by

$$G(s) = \frac{e^{-s/500}}{s + 500}$$

The input to the system is $x(t) = \sin 100\pi t$. In periodic steady state the output of the system is found to be $y(t) = A\sin(100\pi t - \phi)$. The phase angle (ϕ) in degree is

_____.

Q6.14.2 NAT 2014 2M

The loop transfer function of a feedback control system is given by $G(s)H(s) = \dfrac{1}{s(s+1)(9s+1)}$. Its phase crossover frequency (in rad/s), approximated to two decimal places, is

_____.

Q6.12.1 MCQ 2012 1M

A system with transfer function

$$G(s) = \frac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$$

is excited by $\sin(\omega t)$. The steady-state output of the system is zero at

A. $\omega = 1$ rad/s **B.** $\omega = 2$ rad/s
C. $\omega = 3$ rad/s **D.** $\omega = 4$ rad/s

Q6.12.2 MCQ 2012 2M

The open loop transfer function of a unity negative feedback control system is given by

$$G(s) = \frac{150}{s(s+9)(s+25)}$$

The gain margin of the system is

A. 10.8 dB **B.** 22.3 dB **C.** 34.1 dB **D.** 45.6 dB

Common Data Questions

Answer Question Q6.11.1 and Q3.11.2 using the following data.

The open-loop transfer function of a unity negative feedback control system is given by

$$G(s) = \frac{K}{(s+5)^3}$$

The 2nd Question is given in Unit III (Q3.11.2)

Q6.11.1 MCQ 2011 2M

The value of K for the phase margin of the system to be 45° is

A. $250\sqrt{5}$ **B.** $250\sqrt{2}$ **C.** $125\sqrt{5}$ **D.** $125\sqrt{2}$

Q6.09.1 MCQ 2009 1M

A linear time-invariant causal system has a frequency response given in polar form as $\dfrac{1}{\sqrt{1+\omega^2}}\angle -\tan^{-1}\omega$. For input $x(t) = \sin t$, the output is

A. $\dfrac{1}{\sqrt{2}}\cos t$ **B.** $\dfrac{1}{\sqrt{2}}\cos\left(t - \dfrac{\pi}{4}\right)$
C. $\dfrac{1}{\sqrt{2}}\sin t$ **D.** $\dfrac{1}{\sqrt{2}}\sin\left(t - \dfrac{\pi}{4}\right)$

Q6.09.2 MCQ 2009 2M

A unity feedback control loop with an open loop transfer function of the form $\dfrac{K}{s(s+a)}$ has a gain crossover frequency of 1 rad/s and a phase margin of 60°. If an element having a transfer function $\dfrac{s-\sqrt{3}}{s+\sqrt{3}}$ is inserted into the loop, the phase margin will become

A. 0° **B.** 30° **C.** 45° **D.** 60°

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q6.08.1 and Q6.08.2 using the following data. Consider a unity feedback system with open loop transfer function

$$G(s) = \frac{1+6s}{s^2(1+s)(1+2s)}$$

Q6.08.1 MCQ 2008 2M

The phase crossover frequency of the system in radians per second is

A. 0.125 **B.** 0.25 **C.** 0.5 **D.** 1

Q6.08.2 MCQ 2008 2M

The gain margin of the system is

A. 0.125 **B.** 0.25 **C.** 0.5 **D.** 1

Q6.08.3 MCQ 2008 2M

For the closed loop system shown below to be stable, the value of time delay T_D (in seconds) should be less than

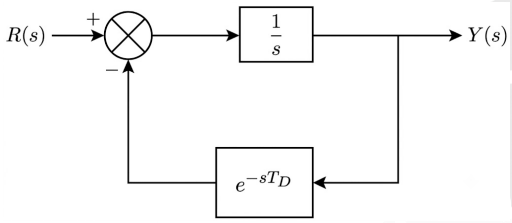


Fig. Q6.08.3

A. $\frac{\pi}{4}$ **B.** $\frac{\pi}{3}$ **C.** $\frac{\pi}{2}$ **D.** π

Q6.06.1 MCQ 2006 2M

A unity feedback system has the following open loop frequency response:

ω (rad/s)	$\lvert G(j\omega)\rvert$	$\angle G(j\omega)$
2	7.5	$-118°$
3	4.8	$-130°$
4	3.15	$-140°$
5	2.25	$-150°$
6	1.70	$-157°$
8	1	$-170°$
10	0.64	$-180°$

The gain and phase margin of the system are

A. 0 dB, $-180°$ **B.** 3.88 dB, $-170°$
C. 0 dB, $10°$ **D.** 3.88 dB, $10°$

Q6.04.1 MCQ 2004 2M

Figure shows the polar plot of a system. The transfer function of the system is

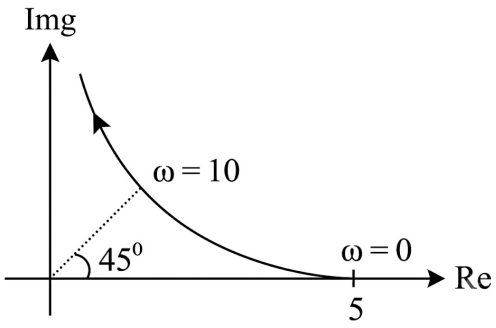


Fig. Q6.04.1

A. $5(1 + 0.1s)$ **B.** $(1 + 0.5s)$
C. $5(1 + 10s)$ **D.** $5(1 + s)$

Q6.04.2 MCQ 2004 2M

The loop transfer function of a system is given by $G(s) = \frac{e^{-0.1s}}{s}$. The phase crossover frequency is given by

A. $\frac{\pi}{2}$ **B.** $\frac{\pi}{10}$ **C.** $\frac{\pi}{0.2}$ **D.** $\frac{\pi}{4}$

Q6.03.1 MCQ 2003 2M

The transfer function of a system is given by

$$\frac{Y(s)}{X(s)} = \frac{e^{-0.1s}}{1+s}$$

If $x(t) = 0.5 \sin t$, then the phase angle between the output and the input will be

A. $-39.27°$ **B.** $-45°$ **C.** $-50.73°$ **D.** $-90°$

Q6.02.1 MCQ 2002 2M

The loop transfer function of a system is given by $G(s)H(s) = \frac{10e^{-Ls}}{s}$. The phase cross-over frequency is 5 rad/s. The value of the dead time L is

A. $\frac{\pi}{20}$ **B.** $\frac{\pi}{10}$ **C.** $-\frac{\pi}{20}$ **D.** zero

Q6.98.1 MCQ 1998 1M

The phase margin of the system for which loop gain $GH(j\omega) = \frac{1}{(j\omega + 1)^3}$ is

A. $-\pi$ **B.** π **C.** 0 **D.** $\pi/2$

Q6.97.1 MCQ 1997 1M

The open-loop transfer function of a feedback control system is $\frac{1}{(s + 1)^3}$. The gain margin of the system is

A. 16 **B.** 8 **C.** 4 **D.** 2

Answer Key

Q6.26.1 0.39 to 0.41	Q6.24.1 A	Q6.23.1 0
Q6.21.1 0.95 to 1.05	Q6.20.1 D	Q6.20.2 65.4 to 65.6
Q6.19.1 D	Q6.19.2 37 to 39	Q6.18.1 90 to 90 or -270 to -270
Q6.18.2 180 to 180 or -180 to -180	Q6.16.1 0.14 to 0.15	Q6.14.1 67 to 69
Q6.14.2 0.30 to 0.34	Q6.12.1 C	Q6.12.2 C
Q6.11.1 B	Q6.09.1 D	Q6.09.2 A
Q6.08.1 C	Q6.08.2 A	Q6.08.3 C
Q6.06.1 D	Q6.04.1 A	Q6.04.2 C
Q6.03.1 C	Q6.02.1 B	Q6.98.1 B
Q6.97.1 B		

UNIT

VII

Nyquist Plot

Topics

1. Nyquist Plot
2. Nyquist Stability Criterion
3. Gain Crossover Frequency (ω_{gc}) from Nyquist Plot
4. Phase Crossover Frequency (ω_{pc}) from Nyquist Plot
5. Gain Margin (GM) using Nyquist Plot
6. Phase Margin (PM) using Nyquist Plot

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 13	**MCQ:** 6	**MSQ:** 0	**NAT:** 7
1M: 5	**2M:** 5	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2025)–2M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q7.25.1 MCQ 2025 1M

Consider the control system block diagram given in Figure (a). The loop transfer function $G(s)H(s)$ does not have any pole on the $j\omega$-axis. The counterclockwise contour with infinite radius, as shown in Figure (b), encircles two poles of $G(s)H(s)$. Choose the correct statement from the following options for closed loop stability of the system.

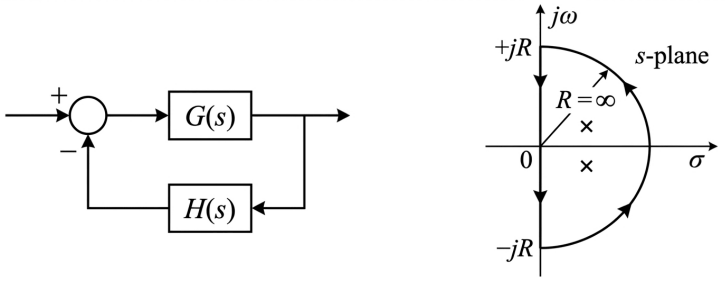


Figure (a) Figure (b)

Fig. Q7.25.1

A. The locus of $G(s)H(s)$ should encircle the origin twice in the counterclockwise direction

B. The locus of $1 + G(s)H(s)$ should encircle the origin twice in the clockwise direction

C. The locus of $G(s)H(s)$ should encircle the $-1 + j0$ point twice in the counterclockwise direction

D. The locus of $1 + G(s)H(s)$ should encircle the $-1 + j0$ point twice in the clockwise direction

Q7.23.1 NAT 2023 1M

The number of times the Nyquist plot of

$$G(s)H(s) = \frac{1}{2}\frac{(s-1)(s-2)}{(s+1)(s+2)}$$

encircles the origin is _____.

Q7.22.1 MCQ 2022 2M

The Nyquist plot of a stable open-loop system $G(j\omega)$ is plotted in the frequency range $0 \le \omega < \infty$ as shown. It is found to intersect a unit circle with center at the origin at the point $P = -0.77 - 0.64j$. The points Q and R lie on $G(j\omega)$ and assume values $Q = 14.40 + 0.00j$ and $R = -0.21 + 0.00j$. The phase margin (PM) and the gain margin (GM) of the system are _____

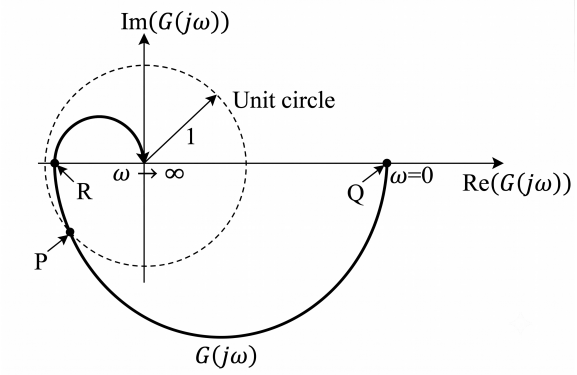


Fig. Q7.22.1

A. PM = 39.7° and GM = 4.76

B. PM = 39.7° and GM = 0.07

C. PM = −39.7° and GM = 4.76

D. PM = −39.7° and GM = 0.07

Q7.21.1 NAT 2021 1M

Taking N as positive for clockwise encirclement, otherwise negative, the number of encirclements N of $(-1, 0)$ in the Nyquist plot of $G(s) = \frac{3}{s-1}$ is _____

Q7.20.1 MCQ 2020 2M

The loop transfer function of a negative feedback system is $G(s)H(s) = \frac{1}{s(s-2)}$. The Nyquist plot for the above system _____

A. encircles $(-1 + j0)$ point once in the clockwise direction

B. encircles $(-1 + j0)$ point once in the counterclockwise direction

C. does not encircle $(-1 + j0)$ point

D. encircles $(-1 + j0)$ point twice in the counterclockwise direction

Q7.18.1 NAT 2018 2M

Consider the standard negative feedback configuration with

$$G(s) = \frac{s^2 + 0.2s + 100}{s^2 - 0.2s + 100} \quad \text{and} \quad H(s) = \frac{1}{2}$$

The number of clockwise encirclements of $(-1, 0)$ in the Nyquist plot of the loop transfer function $G(s)H(s)$ is _____.

Q7.16.1 NAT 2016 1M

The number of times the Nyquist plot of $G(s) = \dfrac{s-1}{s+1}$ will encircle the origin clockwise is

_____.

Q7.15.1 MCQ 2015 2M

A transfer function $G(s)$ with the degree of its numerator polynomial zero and the degree of its denominator polynomial two has a Nyquist plot shown in the figure. The transfer function represents

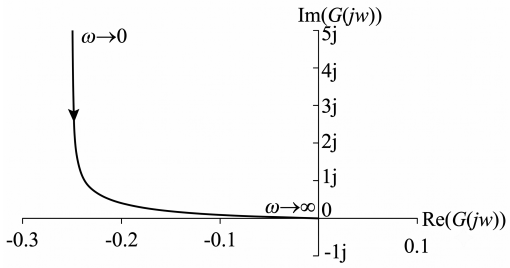


Fig. Q7.15.1

A. a stable, type-0 system

B. a stable, type-1 system

C. an unstable, type-0 system

D. an unstable, type-1 system

Q7.05.1 MCQ 2005 2M

The open loop part of a unity feedback control system $G(s)$ is unstable and has two poles on the right hand side of the s-plane. However, the closed loop system is stable. The number of encirclements made by the Nyquist plot of the $(-1, 0)$ point in the $G(s)$ plane is

A. 2 **B.** 0 **C.** -3 **D.** -1

Q7.02.1 MCQ 2002 1M

The loop transfer function of a system has a pole in the right half of the s-plane. The Nyquist plot makes one clockwise encirclement of $(-1, 0)$ point. The closed loop system is

A. Unstable **B.** Stable
C. Marginally stable **D.** Asymptotically stable

Common Data Questions

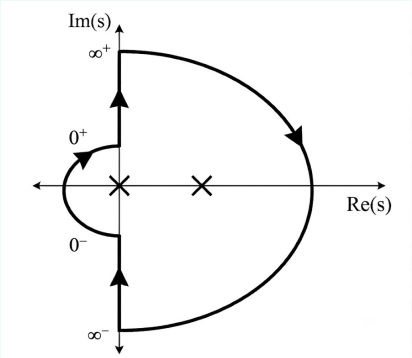
The next 3 questions share the data below.

Answer Questions Q7.01.1, Q7.01.2 and Q7.01.3 using the following data. These three questions together carry a total of **5 marks**.

Consider the open-loop transfer function

$$G(s) = \frac{s+1}{s(s-1)}$$

The Nyquist contour C is shown in the figure. The contour has an indentation to the left of the pole at the origin.



Q7.01.1 NAT 2001

Draw the Nyquist plot for the given system by mapping the contour C.

Q7.01.2 NAT 2001

What is the encirclement criterion for stability of this system with the type of contour chosen?

Q7.01.3 NAT 2001

Using this criterion determine the range of the gain K of the proportional controller over which the closed loop system in unity feedback will be stable.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q7.25.1 B	Q7.23.1 2 to 2	Q7.22.1 A
Q7.21.1 -1 or 1	Q7.20.1 A	Q7.18.1 0 to 0
Q7.16.1 1 to 1	Q7.15.1 D	Q7.05.1 A
Q7.02.1 A	Q7.01.1 *	Q7.01.2 *
Q7.01.3 *		

PrepFusion

UNIT

VIII

Bode Plot

Topics

1. Bode Gain and Phase Plots
2. Minimum and Non-Minimum Phase Systems
3. Bode Plots for Repeated Poles and Zeros
4. Recovery of Transfer Function from Bode Plot
5. Error in Asymptotic Bode Gain Plot
6. Bode Plots of 2nd Order Systems

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 12	**MCQ:** 11	**MSQ:** 0	**NAT:** 1
1M: 5	**2M:** 5	**Descriptive:** 2	**Avg Weightage** ~2M	**Range:** 1M(2018)–2M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q8.23.1 MCQ 2023 2M

The magnitude and phase plots shown in the figure match with the transfer-function _____.

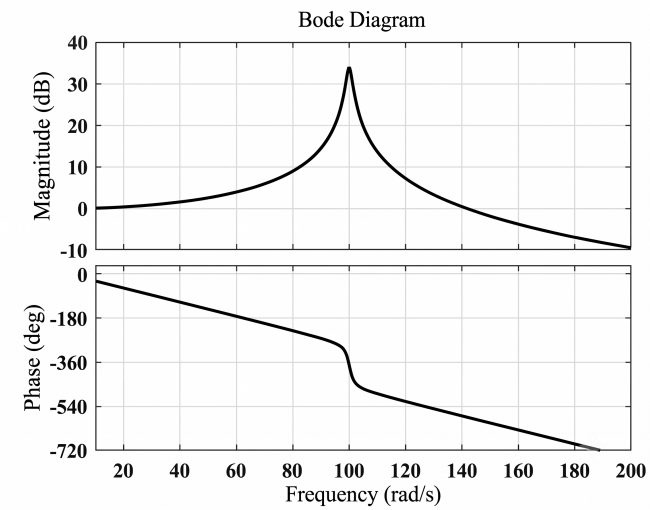


Fig. Q8.23.1

A. $\dfrac{10000}{s^2 + 2s + 10000}$

B. $\dfrac{10000e^{-0.05s}}{s^2 + 2s + 10000}$

C. $\dfrac{10000e^{-0.5\times10^{-12}s}}{s^2 + 2s + 10000}$

D. $\dfrac{100}{s^2 + 2s + 100}$

Q8.19.1 MCQ 2019 2M

The forward path transfer function $L(s)$ of the control system shown in Figure (a) has the asymptotic Bode plot shown in Figure (b). If the disturbance $d(t)$ is given by $d(t) = 0.1\sin(\omega t)$, where $\omega = 5$ rad/s, the steady-state amplitude of the output $y(t)$ is **Figure (a)**

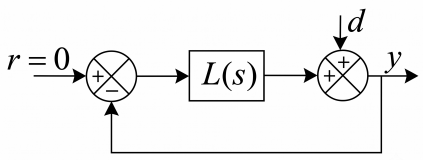


Figure (b)

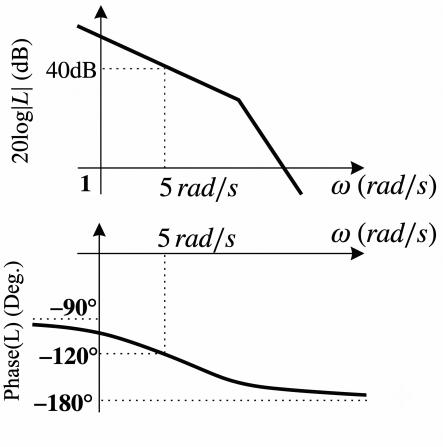


Fig. Q8.19.1

A. 1.00×10^{-3}

B. 2.50×10^{-3}

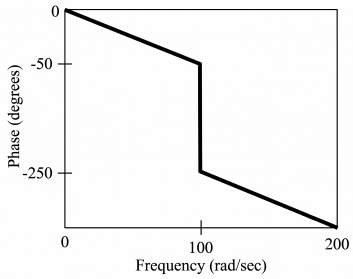
C. 5.00×10^{-3}

D. 10.00×10^{-3}

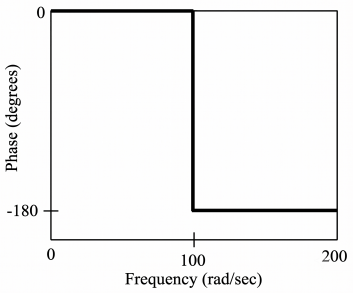
Q8.18.1 MCQ 2018 1M

The approximate phase response of $\dfrac{100^2 e^{-0.01s}}{s^2 + 0.2s + 100^2}$ is

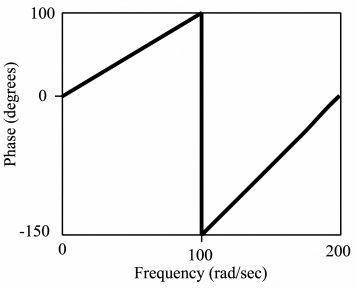
A.



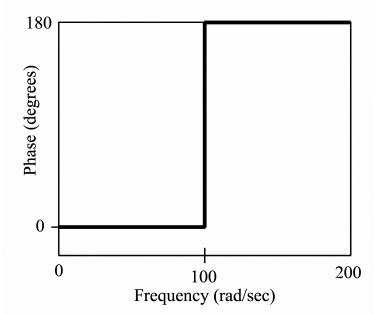
B.



C.



D.



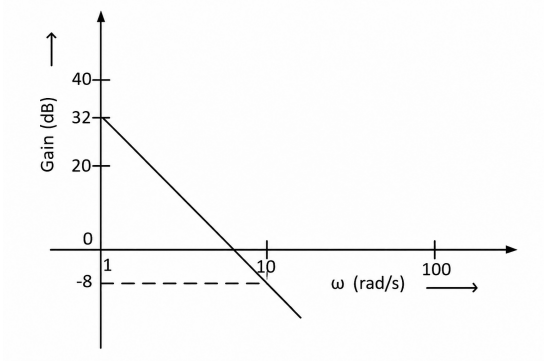


Fig. Q8.13.1

Q8.16.1 MCQ 2016 2M

The transfer transfer function $G(s)$ of a system which has the asymptotic Bode plot shown below is

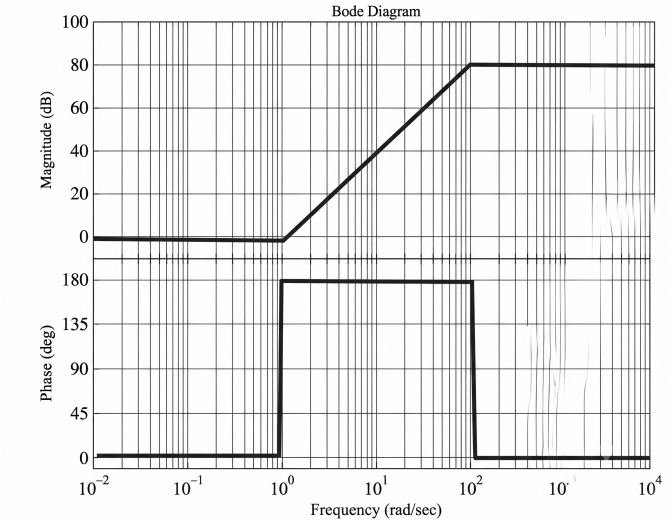


Fig. Q8.16.1

A. $10^4\left(\frac{s-1}{s+100}\right)^2$ **B.** $10^4\left(\frac{s+1}{s+100}\right)^2$

C. $10^4\frac{(s+1)}{(s+100)^2}$ **D.** $10^4\left(\frac{s-1}{s-100}\right)^2$

Q8.13.1 MCQ 2013 1M

The Bode plot of a transfer function $G(s)$ is shown in the figure below.

The gain $(20\log|G(s)|)$ is 32 dB and -8 dB at 1 rad/s and 10 rad/s respectively. The phase is negative for all ω. Then $G(s)$ is

A. $\frac{39.8}{s}$ **B.** $\frac{39.8}{s^2}$ **C.** $\frac{32}{s}$ **D.** $\frac{32}{s^2}$

Q8.08.1 MCQ 2008 1M

The Bode asymptotic plot of a transfer function is given below. In the frequency range shown, the transfer function has

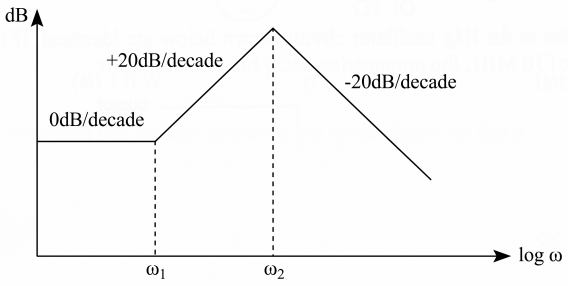


Fig. Q8.08.1

A. 3 poles and 1 zero **B.** 1 pole and 2 zeros
C. 2 poles and 1 zero **D.** 2 poles and 2 zeros

Q8.05.1 MCQ 2005 2M

The asymptotic magnitude Bode plot of an open loop system $G(s)$ with $K > 0$ and all poles and zeros on the left hand side of the s-plane is shown in the figure. It is completely symmetric about ω_c. The minimum absolute phase angle contribution by $G(s)$ is given by

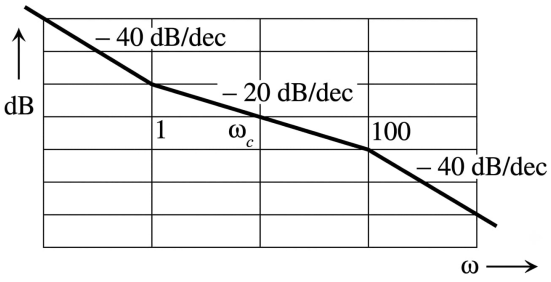


Fig. Q8.05.1

A. 78.6° **B.** 90° **C.** 101.4° **D.** 180°

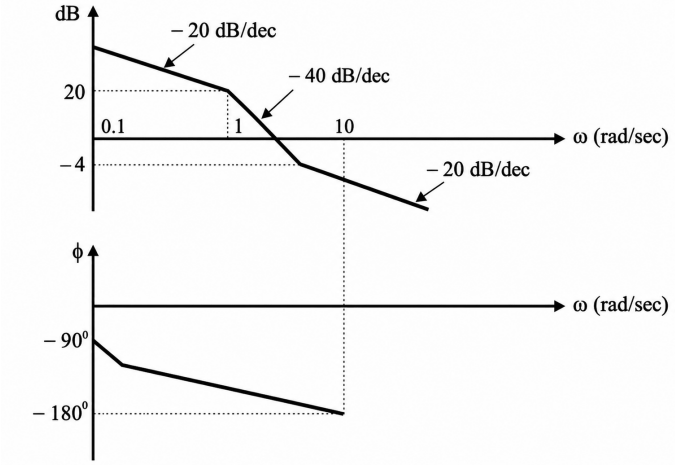


Fig. Q8.02.1

Q8.04.1 MCQ 2004 2M

Figure shows the Bode magnitude plot of a system. The minimum phase transfer function of the system is given by

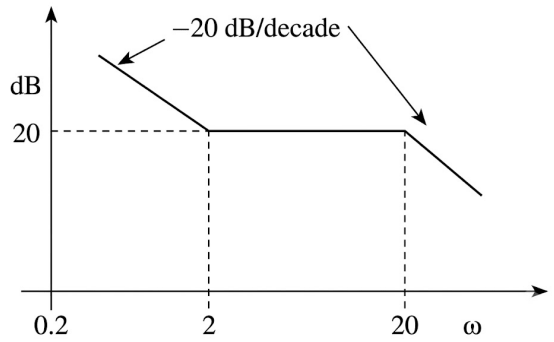


Fig. Q8.04.1

A. $\dfrac{80\left(\frac{s}{2}+1\right)}{s\left(\frac{s}{20}+1\right)}$ **B.** $\dfrac{20\left(\frac{s}{2}+1\right)}{s\left(\frac{s}{20}+1\right)}$

C. $\dfrac{8(s+2)}{s(s+20)}$ **D.** $\dfrac{8(s+2)}{s(s-20)}$

Q8.02.1 NAT 2002 5M

Figure below shows the Bode magnitude and phase plots of a system (not to scale). Determine from the plots the transfer function of the system.

Q8.00.1 MCQ 2000 1M

The asymptotic Bode plot for the gain magnitude of a minimum phase system $G(s)$ is shown below

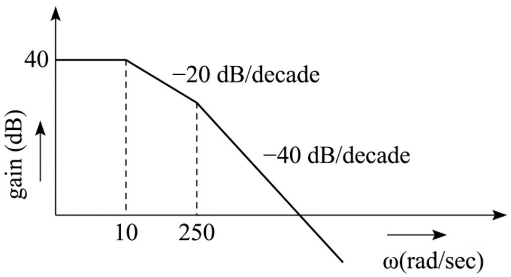


Fig. Q8.00.1

The transfer function is $G(s) =$

A. $\dfrac{100}{(1+s/10)(1+s/250)}$ **B.** $\dfrac{40}{s(s+250)}$

C. $\dfrac{100}{(s+10)(s+250)}$ **D.** $\dfrac{100\,s}{(s+10)(s+250)}$

Q8.96.1 MCQ 1996 1M

Which of the following transfer functions has a minimum phase characteristic?

A. $\dfrac{1+s}{1-s}$ **B.** $\dfrac{1-s}{1+s}$ **C.** $\dfrac{e^{-3s}}{s}$ **D.** $\dfrac{s}{(s+1)(s+2)}$

Q8.92.1 MCQ 1992

The asymptotic approximation to the log-magnitude versus frequency plot (Bode plot) of a unity feedback control system is shown in figure. The system is a minimum phase system. Determine the gain crossover frequency (in rad/sec), Gain margin (in dB) and Phase margin (in degrees).

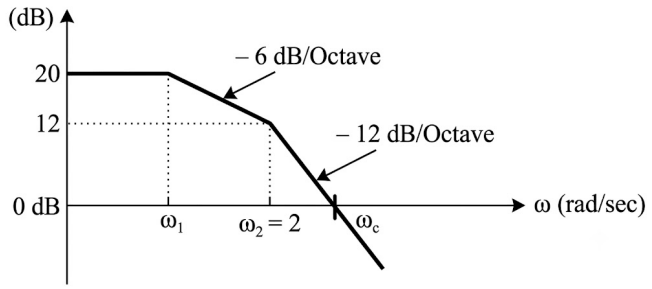


Fig. Q8.92.1

A. 4, 0, ∞
B. 5, ∞, 0
C. 4, ∞, 38°
D. 5, 0, 38°

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q8.23.1 **B**	Q8.19.1 **A**	Q8.18.1 **A**
Q8.16.1 **B**	Q8.13.1 **B**	Q8.08.1 **C**
Q8.05.1 **C**	Q8.04.1 **B**	Q8.02.1 *
Q8.00.1 **A**	Q8.96.1 **D**	Q8.92.1 **C**

UNIT

IX

State Space Analysis

Topics

1. State Variables and State Diagram
2. State Space Representation
3. Eigen Values and Eigen Vectors in State Space
4. Solution of State Space Equations
5. State Transition Matrix and Its Properties
6. Recovery of Transfer Function from State Space
7. Similarity Transforms
8. Controllable and Observable Canonical Forms
9. Parallel and Cascade Decomposition
10. Controllability — Kalman's and Gilbert's Test
11. Observability — Kalman's and Gilbert's Test
12. Observable and Detectable Systems

Unit Snapshot*

Stream: IN	**Questions:** 17	**MCQ:** 15	**MSQ:** 0	**NAT:** 2
1M: 7	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2024)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q9.26.1 MCQ 2026 2M

A linear time invariant (LTI) system is described by:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -10 & -7\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

The poles of the system are _____.

A. $-2, -5$ **B.** $2, 5$ **C.** $-2, 5$ **D.** $2, -5$

Q9.24.1 MCQ 2024 1M

Consider the state-space representation of a system $\dot{x} = Ax + Bu$, where x is the state vector, u is the input, A is the system matrix and B is the input matrix. Choose the matrix A from the following options such that the system has a pole at the origin.

A. $\begin{bmatrix}0 & 1\\ -2 & -3\end{bmatrix}$ **B.** $\begin{bmatrix}1 & -1.5\\ -2 & 3\end{bmatrix}$

C. $\begin{bmatrix}1 & 1.5\\ 2 & -3\end{bmatrix}$ **D.** $\begin{bmatrix}0 & 1\\ -2 & 3\end{bmatrix}$

Q9.22.1 NAT 2022 1M

The transfer function of a system is:

$$\frac{(s+1)(s+3)}{(s+5)(s+7)(s+9)}$$

In the state-space representation of the system, the minimum number of state variables (in integer) necessary is _____.

Q9.19.1 MCQ 2019 2M

The dynamics of the state $\begin{bmatrix}x_1\\ x_2\end{bmatrix}$ of a system is governed by the differential equation

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}1 & 2\\ -3 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}20\\ 10\end{bmatrix}$$

Given that the initial state is $\begin{bmatrix}0\\ 0\end{bmatrix}$, the steady state value of $\begin{bmatrix}x_1\\ x_2\end{bmatrix}$ is

A. $\begin{bmatrix}-30\\ -40\end{bmatrix}$ **B.** $\begin{bmatrix}-20\\ -10\end{bmatrix}$ **C.** $\begin{bmatrix}5\\ -15\end{bmatrix}$ **D.** $\begin{bmatrix}50\\ -35\end{bmatrix}$

Q9.18.1 MCQ 2018 2M

Consider the linear system

$$\dot{x} = \begin{bmatrix}-1 & 0\\ 0 & -2\end{bmatrix}x,$$

with initial condition $x(0) = \begin{bmatrix}1\\ 1\end{bmatrix}$. The solution $x(t)$ for this system is

A. $x(t) = \begin{bmatrix}e^{-t} & te^{-2t}\\ 0 & e^{-2t}\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix}$ **B.** $x(t) = \begin{bmatrix}e^{-t} & 0\\ 0 & e^{2t}\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix}$

C. $x(t) = \begin{bmatrix}e^{-t} & -t^2e^{-2t}\\ 0 & e^{-2t}\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix}$ **D.** $x(t) = \begin{bmatrix}e^{-t} & 0\\ 0 & e^{-2t}\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix}$

Q9.15.1 NAT 2015 2M

A system is represented in state-space as

$$\dot{x} = Ax + Bu,$$

where

$$A = \begin{bmatrix}1 & 2\\ \alpha & 6\end{bmatrix}, \qquad B = \begin{bmatrix}1\\ 1\end{bmatrix}$$

The value of α for which the system is not controllable is _____.

Q9.12.1 MCQ 2012 2M

The state variable description of an LTI system is given by

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & a_1 & 0\\ 0 & 0 & a_2\\ a_3 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

where y is the output and u is the input. The system is controllable for

A. $a_1 \neq 0,\ a_2 = 0,\ a_3 \neq 0$ **B.** $a_1 = 0,\ a_2 \neq 0,\ a_3 \neq 0$

C. $a_1 = 0,\ a_2 \neq 0,\ a_3 = 0$ **D.** $a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0$

Q9.11.1 MCQ 2011 2M

The transfer function of the system described by the state-space equations

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-4 & -1\\ -3 & -1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1\\ 1\end{bmatrix}u,$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

is

A. $\frac{s}{s^2+5s+1}$ **B.** $\frac{2s}{s^2+5s+1}$

C. $\frac{3s}{s^2+5s+1}$ **D.** $\frac{4s}{s^2+5s+1}$

Q9.09.1 MCQ 2009 2M

A linear time-invariant single-input single-output system has a state space model given by

$$\frac{dx}{dt} = Fx + Gu$$

$$y = Hx$$

where

$$F = \begin{bmatrix} 0 & 1 \\ -4 & -2 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Here x is the state vector, u is the input, and y is the output. The damping ratio of the system is

A. 0.25 **B.** 0.5 **C.** 1 **D.** 2

Q9.08.1 MCQ 2008 2M

The state space representation of a system is given by

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

The transfer function $\frac{Y(s)}{U(s)}$ of the system will be

A. $\frac{1}{s}$ **B.** $\frac{1}{s(s+3)}$ **C.** $\frac{1}{s+3}$ **D.** $\frac{1}{s^2}$

Q9.06.1 MCQ 2006 1M

The state-variable representation of a plant is given by

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

where x is the state, u is the input and y is the output. Assuming zero initial condition, the impulse response of the plant is given by

A. $\exp(At)$ **B.** $\int \exp[A(t-\tau)]\, Bu(\tau)\, d\tau$

C. $C\exp(At)B$ **D.** $C\int \exp[A(t-\tau)]\, Bu(\tau)\, d\tau$

Q9.05.1 MCQ 2005 1M

A single input single output linear system is described by the equations

$$\frac{dx}{dt} = Ax + bu$$

$$y = cx + du,$$

where A is an $[n \times n]$ matrix and x and b are $[n \times 1]$ vectors. The transfer function is given by

A. $c[sI - A]^{-1}b$ **B.** $c[sI - A]^{-1}b + d$

C. $c[sI - A]b$ **D.** $c[sI - A]b + d$

Q9.04.1 MCQ 2004 2M

A state space representation for the transfer function $\frac{y(s)}{u(s)} = \frac{s+6}{s^2+5s+6}$ is $\dot{x} = Ax + Bu$, where

$$A = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

and the value of C in $y = Cx$ is

A. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ **B.** $\begin{bmatrix} 6 & 1 \end{bmatrix}$ **C.** $\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & -5 \end{bmatrix}$

Q9.03.1 MCQ 2003 2M

For a dynamic system, various matrices for a state space model are given by

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \text{and } C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

The system poles are located at

A. $-2, -1$ **B.** $-2, -3$ **C.** $-3, -1$ **D.** $-1, 0$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q9.94.1 and Q9.94.2 using the following data.

A first order matrix differential equation of a system is given as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u,$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Q9.94.1 MCQ 1994 1M

The transfer function of the system is

A. $\frac{2}{(s+2)(s+3)}$ **B.** $\frac{2}{(s+1)(s+2)}$

C. $\frac{2}{(s+1)}$ **D.** $\frac{2}{(s+2)}$

Q9.94.2 MCQ 1994 1M

When the input is a unit step and the initial condition of the states is $\begin{bmatrix}0\\1\end{bmatrix}$, then the output is

A. $u(t) - e^{-t}u(t)$ **B.** $e^{-t}u(t)$ **C.** $u(t)$ **D.** $e^{t}u(t)$

Q9.94.3 MCQ 1994 1M

The minimum number of states required to describe the network shown in the figure is:

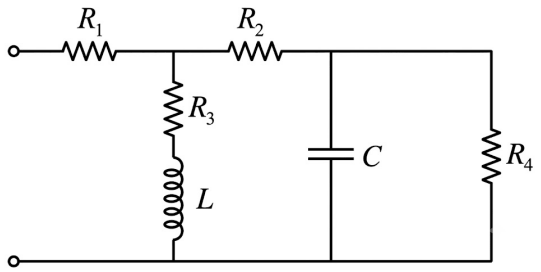


Fig. Q9.94.3

A. 1 **B.** 2 **C.** 3 **D.** 4

Answer Key

Q9.26.1 A	Q9.24.1 B	Q9.22.1 3 to 3
Q9.19.1 D	Q9.18.1 D	Q9.15.1 -3
Q9.12.1 D	Q9.11.1 A	Q9.09.1 B
Q9.08.1 A	Q9.06.1 C	Q9.05.1 B
Q9.04.1 B	Q9.03.1 A	Q9.94.1 D
Q9.94.2 C	Q9.94.3 B	

UNIT

X

Controllers and Compensators

Topics

1. Basics and Types of Controllers — P, PI, PD, PID
2. Lag Compensator
3. Lead Compensator
4. Lag-Lead Compensator
5. Design Problems on Controllers and Compensators

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 42	**MCQ:** 38	**MSQ:** 1	**NAT:** 3
1M: 6	**2M:** 31	**Descriptive:** 5	**Avg Weightage** ~3M	**Range:** 1M(2020)–4M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q10.25.1 MCQ 2025 2M

The figure shows a closed-loop system with a plant $G(s) = \frac{1}{s^2}$ and a lead compensator $C(s)$. The compensator is designed to place the dominant closed-loop poles at $-1.5 \pm j\frac{\sqrt{27}}{2}$. From the following options, choose the phase lead that the compensator needs to contribute.

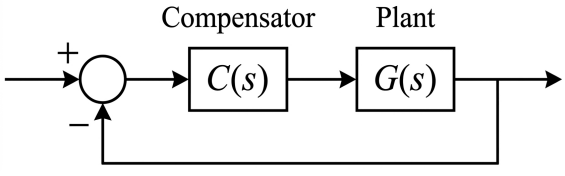


Fig. Q10.25.1

A. 30° **B.** 60° **C.** 90° **D.** 120°

Q10.25.2 MSQ 2025 2M

The plant in the feedback control system shown in the figure is $P(s) = \frac{a}{s^2 - b^2}$, where $a > 0$ and $b > 0$. The type(s) of controller $C(s)$ that CANNOT stabilize the plant is/are.

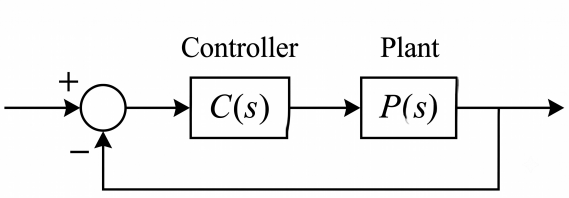


Fig. Q10.25.2

- ☐ **A.** proportional (P) controller
- ☐ **B.** integral (I) controller
- ☐ **C.** proportional-integral (PI) controller
- ☐ **D.** proportional-derivative (PD) controller

Q10.23.1 NAT 2023 2M

Consider a standard negative feedback configuration with $G(s) = \frac{1}{(s-2)(s-3)}$ and the controller $C(s) = K_P + \frac{K_I}{s} + K_D s$. The root-locus of $G(s)C(s)$ is presented in the figure below. The gain $C(j\omega) = 2$ at $\omega = 1$ rad/s. The value of K_D is

_____**(rounded off to one decimal place)**.

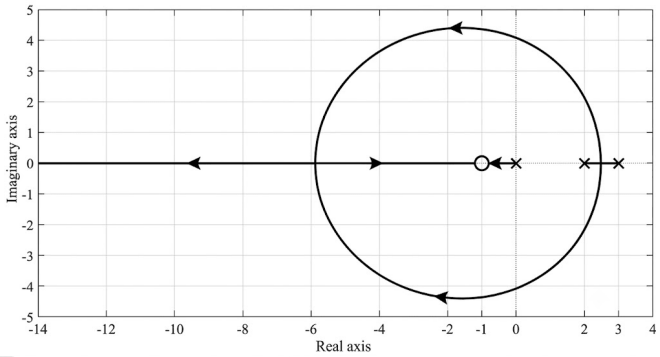


Fig. Q10.23.1

Q10.22.1 MCQ 2022 2M

A proportional-integral-derivative (PID) controller is employed to stably control a plant with transfer function $P(s) = \frac{1}{(s+1)(s+2)}$. Now, the proportional gain is increased by a factor of 2, the integral gain is increased by a factor of 3, and the derivative gain is left unchanged. Given that the closed-loop system continues to remain stable with the new gains, the steady-state error in tracking a ramp reference signal _____

A. Remains unchanged

B. Decreases by a factor of 2

C. Decreases by a factor of 3

D. Decreases by a factor of 5

Q10.22.2 NAT 2022 2M

In a unity-gain feedback control system, the plant

$$P(s) = \frac{0.001}{s(2s+1)(0.01s+1)}$$

is controlled by a lag compensator $C(s) = \frac{s+10}{s+0.1}$. The slope (in dB/decade) of the asymptotic Bode magnitude plot of the loop gain at $\omega = 3$ rad/s is _____ (in integer).

Q10.21.1 MCQ 2021 2M

Consider a unity feedback configuration with a plant and a PID controller as shown in the figure.

$$G(s) = \frac{1}{(s+1)(s+3)}$$

$$C(s) = K\frac{(s+3-j)(s+3+j)}{s}$$

with K being scalar. The closed loop is

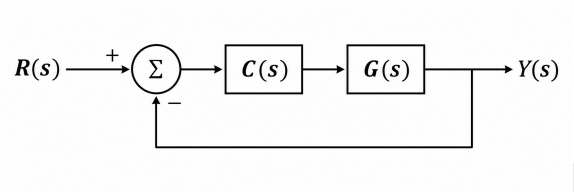


Fig. Q10.21.1

A. only stable for $K > 0$

B. only stable for K between -1 and $+1$

C. only stable for $K < 0$

D. stable for all values of K

Q10.20.1 MCQ 2020 1M

A phase lead network has the transfer function

$$G(s) = \frac{1+0.2s}{1+0.05s}$$

The angular frequency at which the maximum phase shift for the network occurs is _____

A. 10 rad/s **B.** 20 rad/s **C.** 100 rad/s **D.** 200 rad/s

Q10.19.1 MCQ 2019 2M

In the control system shown in the figure below, a reference signal $r(t) = t^2$ is applied at time $t = 0$. The control system employs a PID controller $C(s) = K_P + \frac{K_I}{s} + K_D s$ and the plant has a transfer function $P(s) = \frac{3}{s}$. If $K_P = 10$, $K_I = 1$ and $K_D = 2$, the steady state value of e is

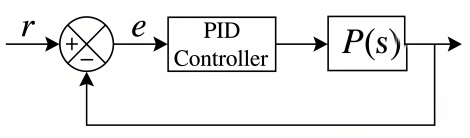


Fig. Q10.19.1

A. 0 **B.** 2/3 **C.** 1 **D.** ∞

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.13.1 and Q10.13.2 using the following data.

Consider a plant with the transfer function $G(s) = \frac{1}{(s+1)^3}$. Let K_u and T_u be the ultimate gain and ultimate period corresponding to the frequency response based closed loop Ziegler-Nichols cycling method, respectively. The Ziegler-Nichols tuning rule for a P-controller is given as: $K = 0.5K_u$.

Q10.13.1 MCQ 2013 2M

The values of K_u and T_u, respectively, are

A. $2\sqrt{2}$ and 2π **B.** 8 and 2π

C. 8 and $\frac{2\pi}{\sqrt{3}}$ **D.** $2\sqrt{2}$ and $\frac{2\pi}{\sqrt{3}}$

Q10.13.2 MCQ 2013 2M

The gain of the transfer function between the plant output and an additive load disturbance input of frequency $\frac{2\pi}{T_u}$ in closed loop with a P-controller designed according to the Ziegler-Nichols tuning rule as given above is

A. −1.0 **B.** 0.5 **C.** 1.0 **D.** 2.0

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.12.1 and Q10.12.2 using the following data.

The transfer function of a compensator is given as

$$G_c(s) = \frac{s+a}{s+b}.$$

Q10.12.1 MCQ 2012 2M

$G_c(s)$ is a lead compensator if

A. $a = 1,\ b = 2$ **B.** $a = 3,\ b = 2$

C. $a = -3,\ b = -1$ **D.** $a = 3,\ b = 1$

Q10.12.2 MCQ 2012 2M

The phase of the above lead compensator is maximum at

A. $\sqrt{2}$ rad/s **B.** $\sqrt{3}$ rad/s **C.** $\sqrt{6}$ rad/s **D.** $1/\sqrt{3}$ rad/s

Q10.10.1 MCQ 2010 2M

The asymptotic Bode magnitude plot of a lead network with its pole and zero on the left half of the s-plane is shown in the figure. The frequency at which the phase angle of the network is maximum (in rad/s) is

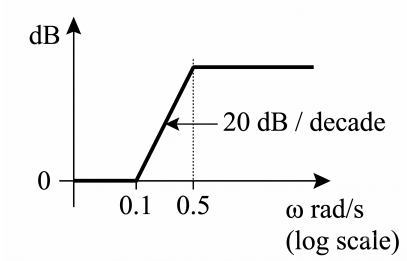


Fig. Q10.10.1

A. $\frac{3}{\sqrt{10}}$ **B.** $\frac{1}{\sqrt{20}}$ **C.** $\frac{1}{20}$ **D.** $\frac{1}{30}$

Q10.09.1 MCQ 2009 1M

A plant with a transfer function $\frac{2}{s(s+3)}$ is controlled by a PI controller with $K_p = 1$ and $K_i \geq 0$ in a unity feedback configuration. The lowest value of K_i that ensures zero steady state error for a step change in the reference input is

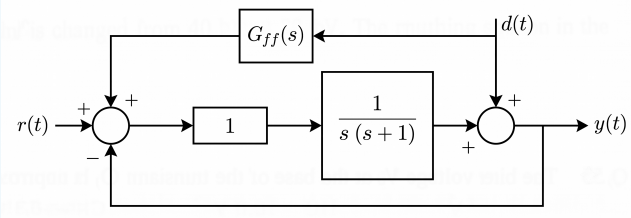
A. 0 **B.** $\frac{1}{3}$ **C.** $\frac{1}{2}$ **D.** 1

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.09.2 and Q10.09.3 using the following data.

A disturbance input $d(t)$ is injected into the unity feedback control loop shown in the figure. Take the reference input $r(t)$ to be a unit step.



Q10.09.2 MCQ 2009 2M

If the disturbance is measurable, its effect on the output can be minimized significantly using a feedforward controller $G_{ff}(s)$. To eliminate the component of the output due to $d(t) = \sin t$, $G_{ff}(j\omega)\big|_{\omega=1}$ should be

A. $\frac{1}{\sqrt{2}}\angle -\frac{3\pi}{4}$ **B.** $\frac{1}{\sqrt{2}}\angle \frac{\pi}{4}$ **C.** $\sqrt{2}\angle \pi$ **D.** $\sqrt{2}\angle -\frac{\pi}{4}$

Q10.09.3 MCQ 2009 2M

Let $G_{ff}(s)$ be a PD controller. If $d(t) = \sin 2t$, the amplitude of the frequency component of $y(t)$ due to $d(t)$ is

A. $\sqrt{\frac{5}{13}}$ **B.** $\sqrt{\frac{9}{13}}$ **C.** $\sqrt{\frac{17}{13}}$ **D.** $\sqrt{\frac{20}{13}}$

Q10.07.1 MCQ 2007 2M

A cascade control system with proportional controllers is shown below.

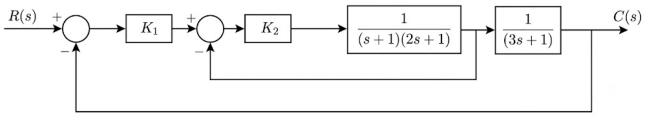


Fig. Q10.07.1

Theoretically, the largest values of the gains K_1 and K_2 that can be set without causing instability of the closed loop system are

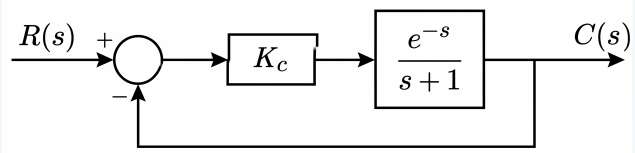
A. 10 and 100 **B.** 100 and 10
C. 10 and 10 **D.** ∞ and ∞

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.07.2 and Q10.07.3 using the following data.

The following figure represents a proportional control scheme of a first order system with transportation lag.



Q10.07.2 MCQ 2007 2M

The angular frequency in radians/s at which the loop phase lag becomes 180° is

A. 0.408 **B.** 0.818 **C.** 1.56 **D.** 2.03

Q10.07.3 MCQ 2007 2M

The steady state error for a unit step input when the gain $K_c = 1$ is

A. $\frac{1}{4}$ **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Q10.06.1 MCQ 2006 2M

The transfer function of a position servo system is given as

$$G(s) = \frac{1}{s(s+1)}.$$

A first order compensator is designed in a unity feedback configuration so that the poles of the compensated system are placed at $-1 \pm j1$ and -4. The transfer function of the compensator is

A. $\frac{s+3}{2(s+5)}$ **B.** $\frac{2s+3}{s+5}$ **C.** $\frac{5(s+1.6)}{s+5}$ **D.** $\frac{3(2s+3)}{s+4}$

Q10.06.2 MCQ 2006 2M

The range of the controller gains $(K_p,\ K_I)$ that makes the closed loop control system (shown in the following figure) stable is given as

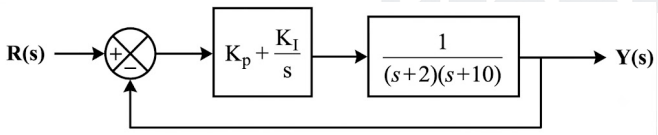


Fig. Q10.06.2

A. $K_I < 0$ and $K_p < \frac{K_I}{12} - 20$

B. $K_I > 0$ and $K_p > 0$

C. $K_I < 0$ and $K_p > \frac{K_I}{12} - 20$

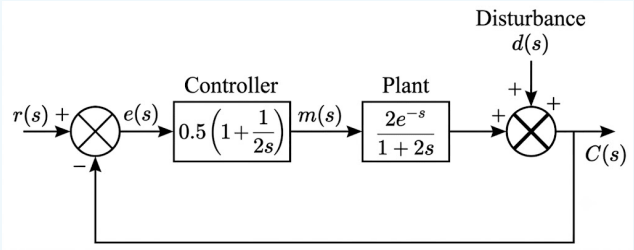
D. $K_I > 0$ and $K_p > \frac{K_I}{12} - 20$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.06.3 and Q10.06.4 using the following data.

The following figure describes the block diagram of a closed loop process control system. The unit of time is given in minute.



Q10.06.3 MCQ 2006 2M

Suppose a disturbance signal $d(t) = \sin 0.2t$ unit is applied. Then at steady state, the amplitude of the output $c(t)$ due to the effect of disturbance alone is

A. 0.129 unit **B.** 0.40 unit **C.** 0.528 unit **D.** 2.102 unit

Q10.06.4 MCQ 2006 2M

The control action recommended for reducing the effect of disturbance at the output (provided that the disturbance is measurable) is

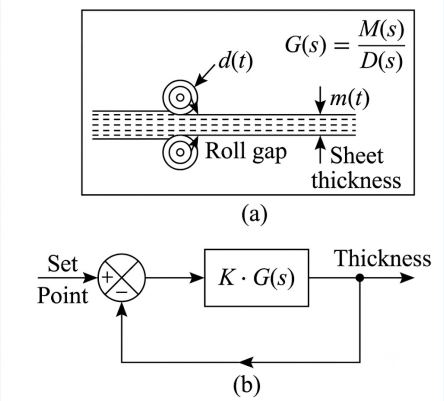
A. cascade control **B.** P–D control
C. ratio control **D.** feedback-feedforward control

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.05.1 and Q10.05.2 using the following data.

The thickness of a metal sheet from a rolling mill is manipulated by controlling the roll gap $d(t)$ and measuring the sheet thickness $m(t)$ at a certain distance from the rolls as shown in the figure. The thickness of the sheet at the roll out point is equal to the roll gap $d(t)$ and remains invariant during transit.



Q10.05.1 MCQ 2005 2M

Given the distance at which the thickness is measured is 0.5 m from the roll gap and the velocity of the sheet 0.2 m/s, the transfer function system, $G(s) = \frac{M(s)}{D(s)}$, is

A. $e^{-2.5s}$ **B.** $e^{+2.5s}$ **C.** $e^{+0.4s}$ **D.** $e^{-0.4s}$

Q10.05.2 MCQ 2005 2M

The gap $d(t)$ is controlled by a proportional controller along with a hydraulic actuator of combined gain K and a unity feedback loop exists around $K \cdot G(s)$ (Fig. b). Increasing K beyond a certain value causes the system to become unstable. The frequency of oscillations at the critical value of K is

A. $\omega = \pi$ **B.** $\omega = 2.5\pi$ **C.** $\omega = 0.4\pi$ **D.** $\omega = \frac{\pi}{2.5}$

Q10.04.1 MCQ 2004 1M

The proportional band of a PI controller is 50% and the reset time is 0.2 s. The transfer function of the controller is given by

A. $50\left(1 + \frac{1}{0.2s}\right)$ **B.** $50\left(1 + \frac{1}{5s}\right)$

C. $2\left(1 + \frac{1}{0.2s}\right)$ **D.** $2\left(1 + \frac{1}{s}\right)$

Q10.03.1 MCQ 2003 2M

A PID controller has the transfer function $2 + \frac{0.4}{s}$ with the unit of time expressed in minutes. The parameters proportional band and reset time for the above controller are respectively

A. 200% and 0.4 min. **B.** 50% and 0.4 min.
C. 200% and 5 min. **D.** 50% and 5 min.

Q10.03.2 MCQ 2003 2M

In figure K_1 and K_2 are proportional controllers. For the control system to remain stable, the upper limits of K_1 and K_2 are

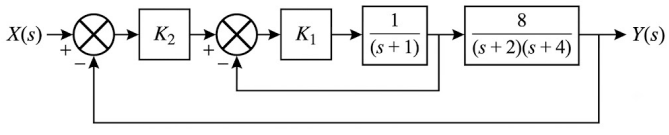


Fig. Q10.03.2

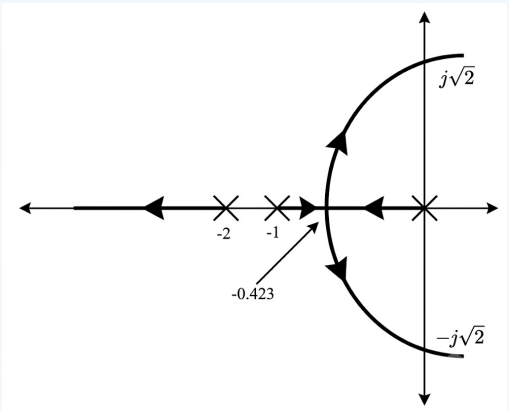
A. ∞, ∞ **B.** $1, \infty$ **C.** $\infty, 10$ **D.** $10, 1$

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.03.3 and Q10.03.4 using the following data.

The root locus plot for a system, with transfer function $\frac{2}{s(s+1)(s+2)}$, is shown in figure below. A unity feedback proportional control system is built using this system.



Q10.03.3 MCQ 2003 2M

The maximum possible controller gain, for which the unity feedback system is stable, is approximately

A. 6.0 **B.** 3.0 **C.** 0.4 **D.** 0.2

Q10.03.4 MCQ 2003 2M

The maximum possible controller gain, for which the unity feedback system exhibits a non-oscillatory response to a unit step input, is

A. 6.0 **B.** 3.0 **C.** 0.4 **D.** 0.2

Q10.02.1 MCQ 2002 1M

An integral control is used to

A. Improve the transient response

B. Reduce the offset

C. Eliminate the offset

D. Reduce the settling time

Q10.02.2 MCQ 2002 2M

The transfer function of a PID Controller is given by

$$G(s) = 4\left(1 + \frac{1}{2s} + 0.5s\right),$$

as ω tends to infinity

A. Magnitude of $G(j\omega)$ tends to zero and phase angle of $G(j\omega)$ tends to infinity

B. Magnitude of $G(j\omega)$ tends to infinity and phase angle of $G(j\omega)$ tends to zero

C. Magnitude of $G(j\omega)$ tends to infinity and phase angle of $G(j\omega)$ tends to $+90°$

D. Magnitude of $G(j\omega)$ tends to zero and phase angle of $G(j\omega)$ tends to $+90°$

Q10.01.1 MCQ 2001 2M

A process in a feedback loop with a proportional controller with ultimate gain $K_u = 10$ is oscillating at a frequency of $P_u = 8$ Hz. The Ziegler-Nichols setting for the proportional controller is

A. 8 **B.** 5 **C.** 1.25 **D.** 10

Q10.00.1 MCQ 2000 2M

The proportional gain of a PID controller can be expressed in terms of its proportional band (PB) as

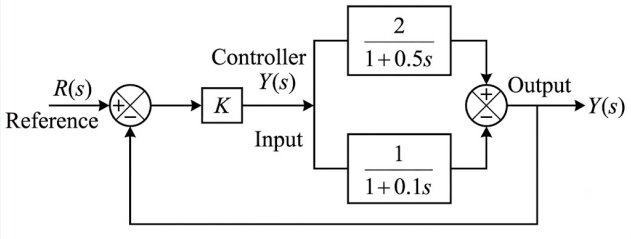
A. PB **B.** $\frac{100}{PB}$ **C.** $\frac{PB}{100}$ **D.** None

Common Data Questions

The next 2 questions share the data below.

Answer Questions Q10.00.2 and Q10.00.3 using the following data. These two questions together carry a total of **5 marks**.

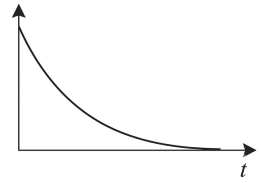
Consider the plant shown in the below figure with a proportional controller with a positive gain.



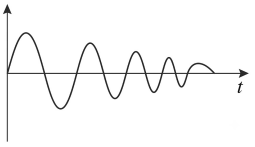
Q10.00.2 MCQ 2000

The open loop step response of the plant is:

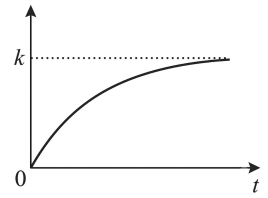
A.



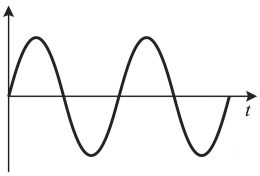
B.



C.



D.



Q10.00.3 NAT 2000

For stable operation, the steady state gain of the above closed loop controlled system, is limited to _____

Q10.99.1 MCQ 1999

A first order system with a proportional controller in the negative feedback loop has an offset to a step input. This offset can be eliminated by

A. Adding a derivative mode to the controller.

B. Adding an integral mode to the controller.

C. Decreasing the magnitude of the gain of the proportional controller.

D. Adding a delay in the controller loop.

Q10.98.1 MCQ 1998

The input error voltage to a PD-Controller, $e(t)$ is shown in the figure. The output voltage of the controller $e_0(t)$ is

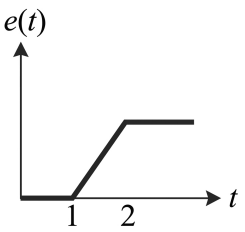
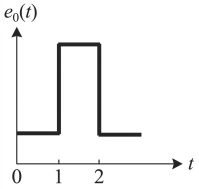
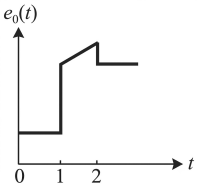
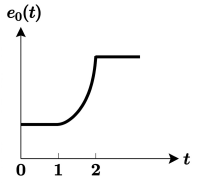
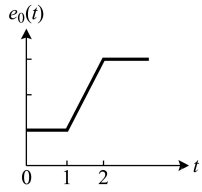


Fig. Q10.98.1

A.


B.


C.


D.


Q10.97.1 MCQ 1997 1M

The transfer function of a basic PD controller is given by (K's are real constants)

A. $K_0 + \frac{K_1}{s} + K_2 s$

B. $K_0 + K_2 s$

C. $K_1 s + K_2 s$

D. $K_0 + \frac{K_1}{s}$

Q10.94.1 MCQ 1994

A cascade controller is used when the process

A. Gain is too small

B. Gain is too large

C. Has widely different time constants

D. Oscillation at the output is not permitted

Q10.93.1 MCQ 1993 2M

A plant is controlled by a proportional controller. If a time delay element is introduced in the loop then

A. gain margin remains the same

B. phase margin remains the same

C. gain margin decreases

D. phase margin decreases

Q10.92.1 MCQ 1992 1M

A temperature control system is usually very sluggish. To improve its dynamics

A. a PI controller can be used

B. an I controller can be used

C. a PID controller with large I and a negligible D action can be used

D. a PD controller can be used

Answer Key

Note: **MTA** Marks To All awarded — see detailed solutions.

Q10.25.1 **B**	Q10.25.2 **A, B, C**	Q10.23.1 **0.9 to 1.1**
Q10.22.1 **C**	Q10.22.2 **-60**	Q10.21.1 **A**
Q10.20.1 **A**	Q10.19.1 **B**	Q10.13.1 **C**
Q10.13.2 **D**	Q10.12.1 **A**	Q10.12.2 **A**
Q10.10.1 **B**	Q10.09.1 **A**	Q10.09.2 **D**
Q10.09.3 **B**	Q10.07.1 **MTA**	Q10.07.2 **D**
Q10.07.3 **B**	Q10.06.1 **C**	Q10.06.2 **D**
Q10.06.3 **B**	Q10.06.4 **D**	Q10.05.1 **A**
Q10.05.2 **C, D**	Q10.04.1 **C**	Q10.03.1 **D**
Q10.03.2 **A**	Q10.03.3 **B**	Q10.03.4 **D**
Q10.02.1 **C**	Q10.02.2 **C**	Q10.01.1 **B**
Q10.00.1 **B**	Q10.00.2 **C**	Q10.00.3 **0.67**
Q10.99.1 **B**	Q10.98.1 **B**	Q10.97.1 **B**
Q10.94.1 **C**	Q10.93.1 **D**	Q10.92.1 **D**

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PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

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